

Lecture 8A: Visual Imitation

Jitendra Malik

Towards an Embodied AI, inspired by child development



Perception and Interaction

Language

456

A. M. TURING :

Instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child's? If this were then subjected to an appropriate course of education one would obtain the adult brain. Presumably the child-brain is something like a note-book as one buys it from the stationers. Rather little mechanism, and lots of blank sheets. (Mechanism and writing are from our point of view almost synonymous.) Our hope is that there is so little mechanism in the child-brain that something like it can be easily programmed. The amount of work in the education we can assume, as a first approximation, to be much the same as for the human child.

Turing (1950)
Computing Machinery
And Intelligence

The Development of Embodied Cognition: Six Lessons from Babies

Linda Smith & Michael Gasser

- Be multi-modal
- Be incremental
- Be physical
- Explore
- Be social
- Use language

SFV: Reinforcement Learning of Skills from Videos

Xue Bin Peng, Angjoo Kanazawa, Jitendra Malik, Pieter Abbeel, Sergey Levine
SIGGRAPH Asia 2018



Xue Bin Peng



Angjoo Kanazawa



Jitendra Malik



Pieter Abbeel



Sergey Levine

Instead... use video



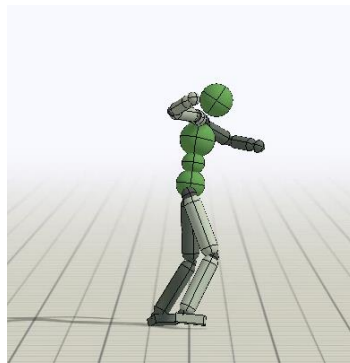
Deep RL based motion imitation

DeepMimic [Peng et al. SIGGRAPH 2018]



Character
In simulation

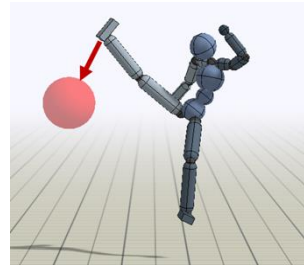
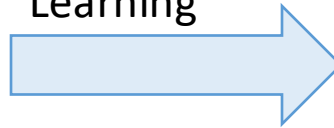
Need
Actions



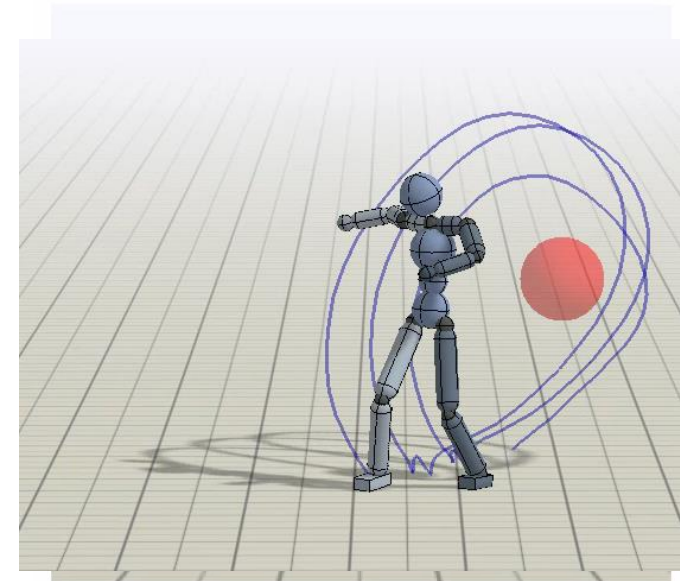
Reference Motion
(**Mocap**)

Only Recorded
joint angles

Reinforcement
Learning



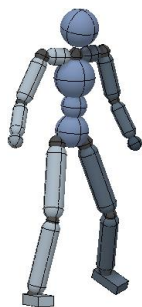
Task: Hit Target



Still requires reference motion from mocap

Deep RL based motion imitation

DeepMimic [Peng et al. SIGGRAPH 2018]



Character
In simulation

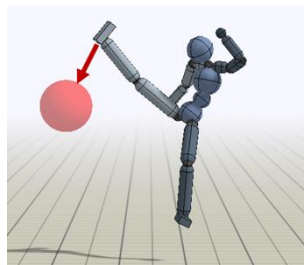
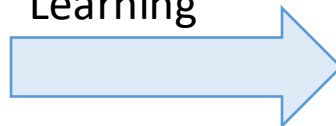
Need
Actions

Replace with
HMR
[Kanazawa et al.
CVPR'18]

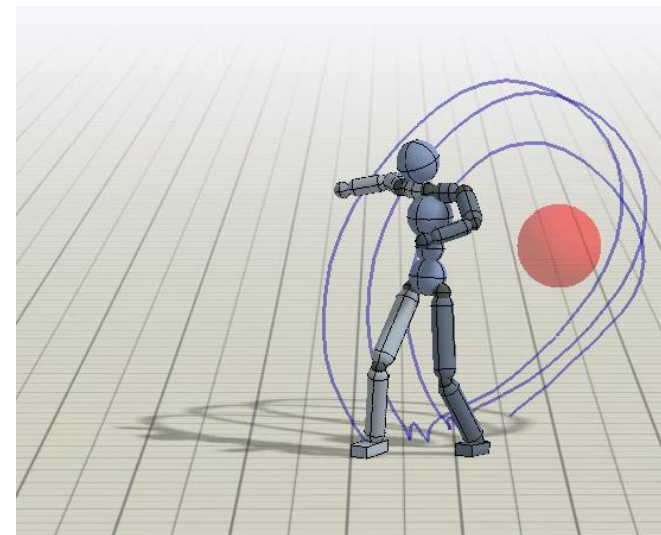
Reference Motion
(**Mocap**)

Only Recorded
joint angles

Reinforcement
Learning



Task: Hit Target



Learning Dynamic Skills from Videos



Video

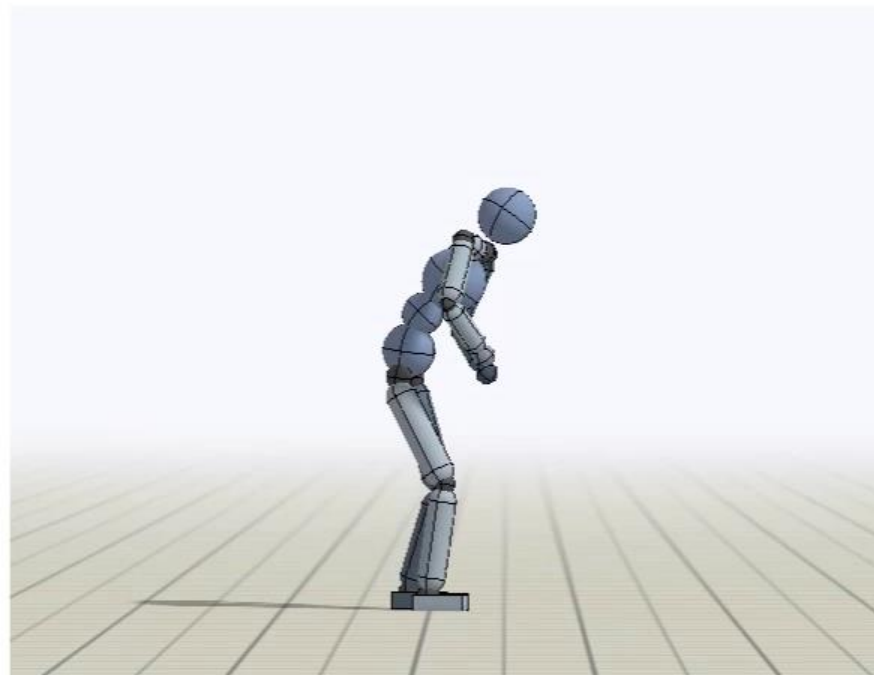


HMR + temporal smoothing

Use recovered 3D pose to train a physically simulated agent

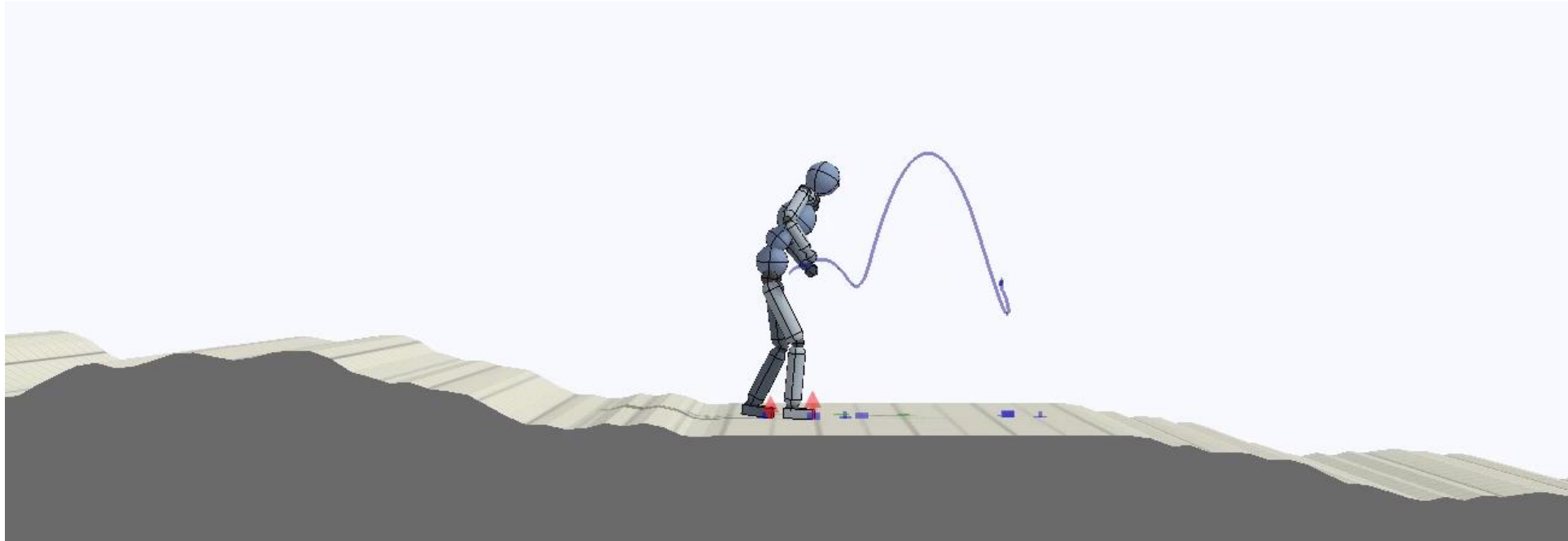


Video



Simulation

Environment Retargeting



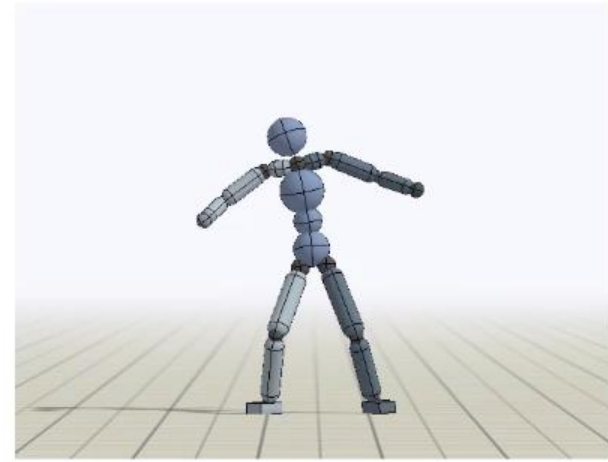
Humanoid: Cartwheel



Video: Cartwheel B

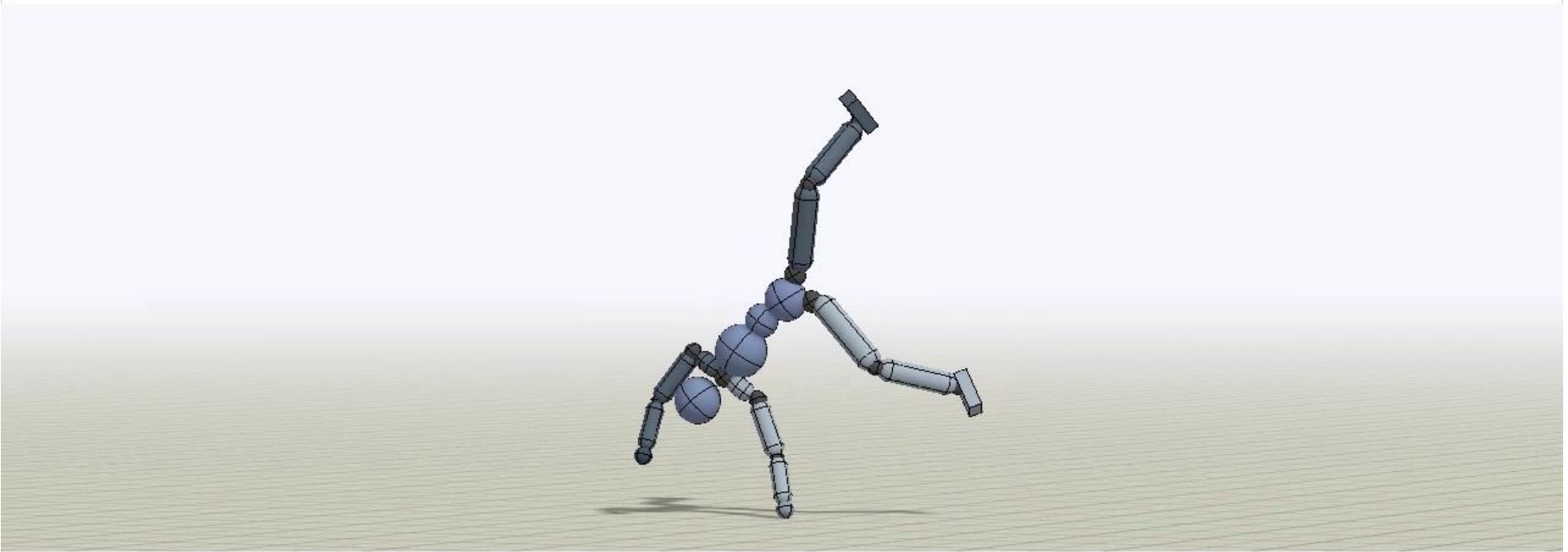


Reference Motion



Policy

Robustness



The policies learn to recover from significant perturbations.

Humanoid: Frontflip



Video: Frontflip

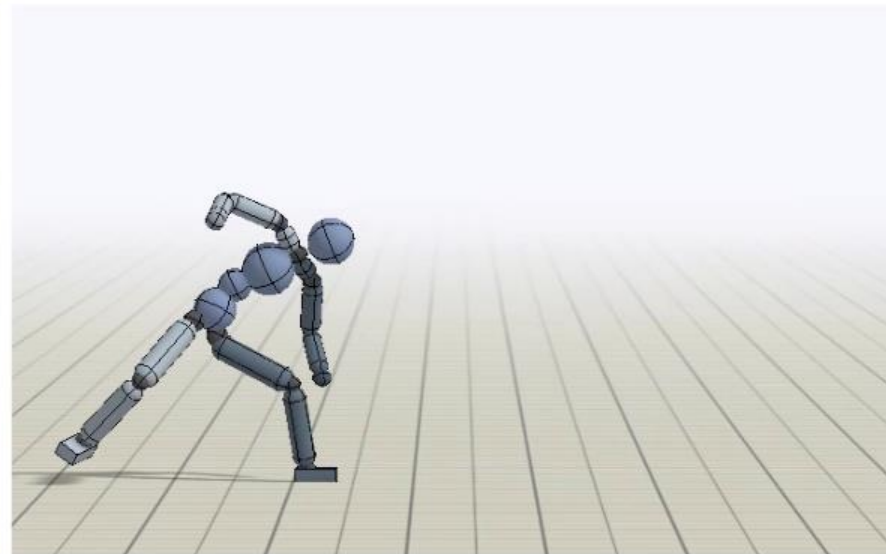


Policy

Humanoid: Spin



Video: Spin

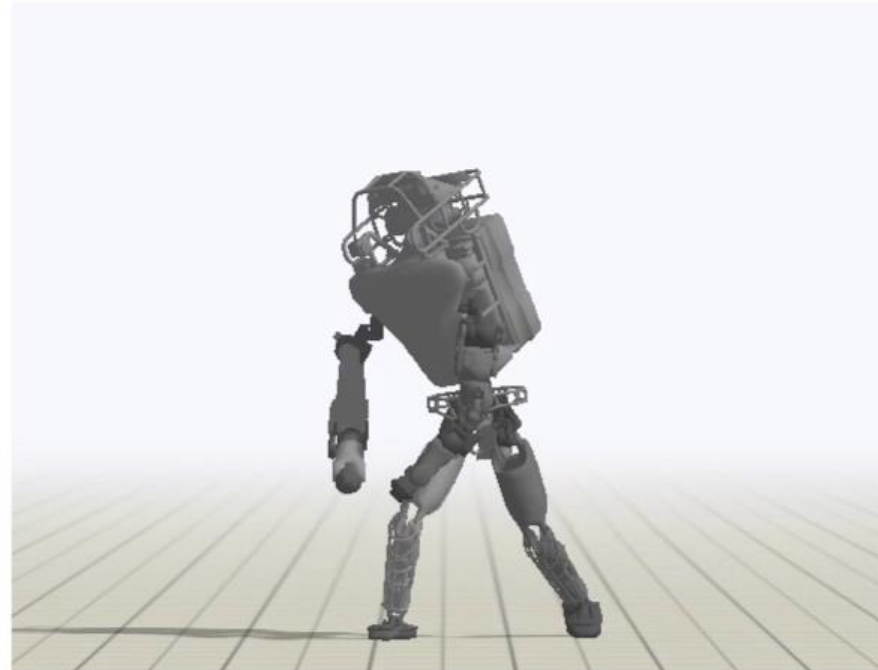


Policy

Atlas: Dance

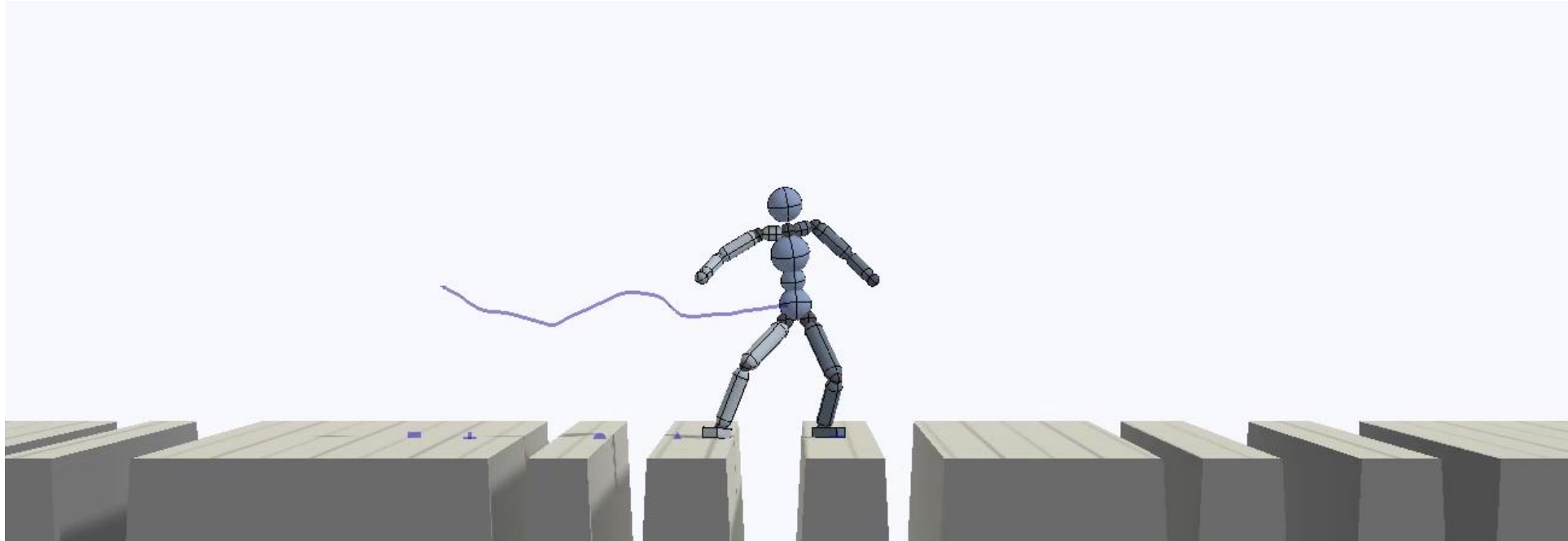


Video: Dance



Policy

Environment Retargeting



Reconstructing 4D Human Actions

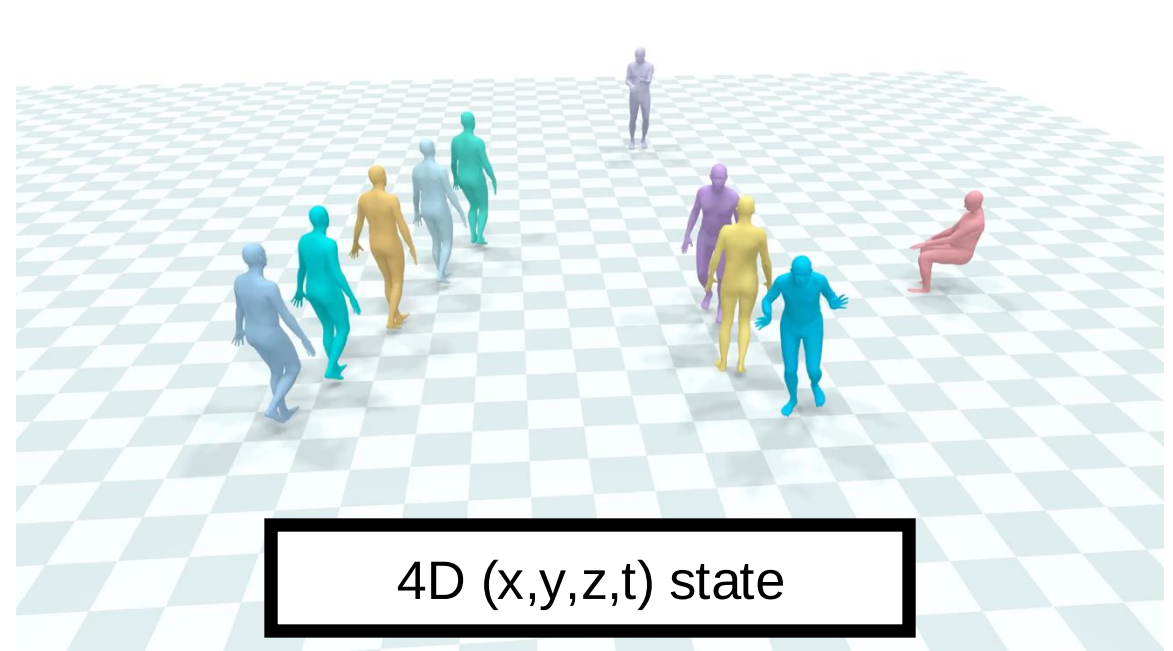
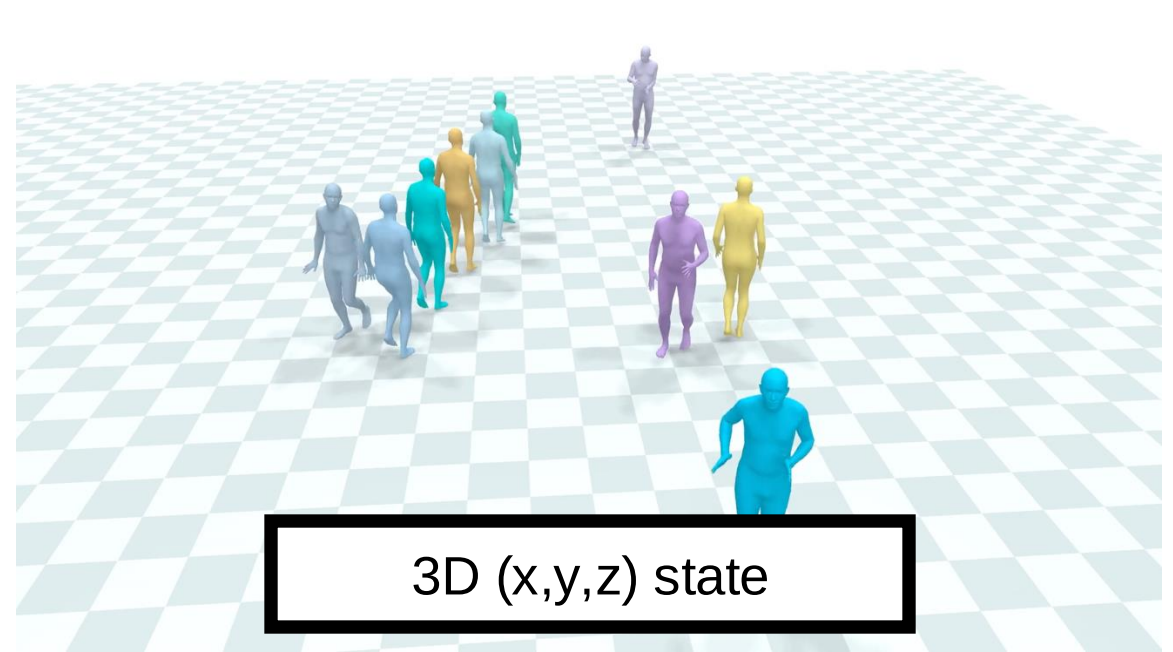


Jitendra Malik

UC Berkeley

FAIR, Meta





Humans in 4D: Reconstructing and Tracking Humans with Transformers

ICCV 2023

Shubham Goel

Georgios Pavlakos

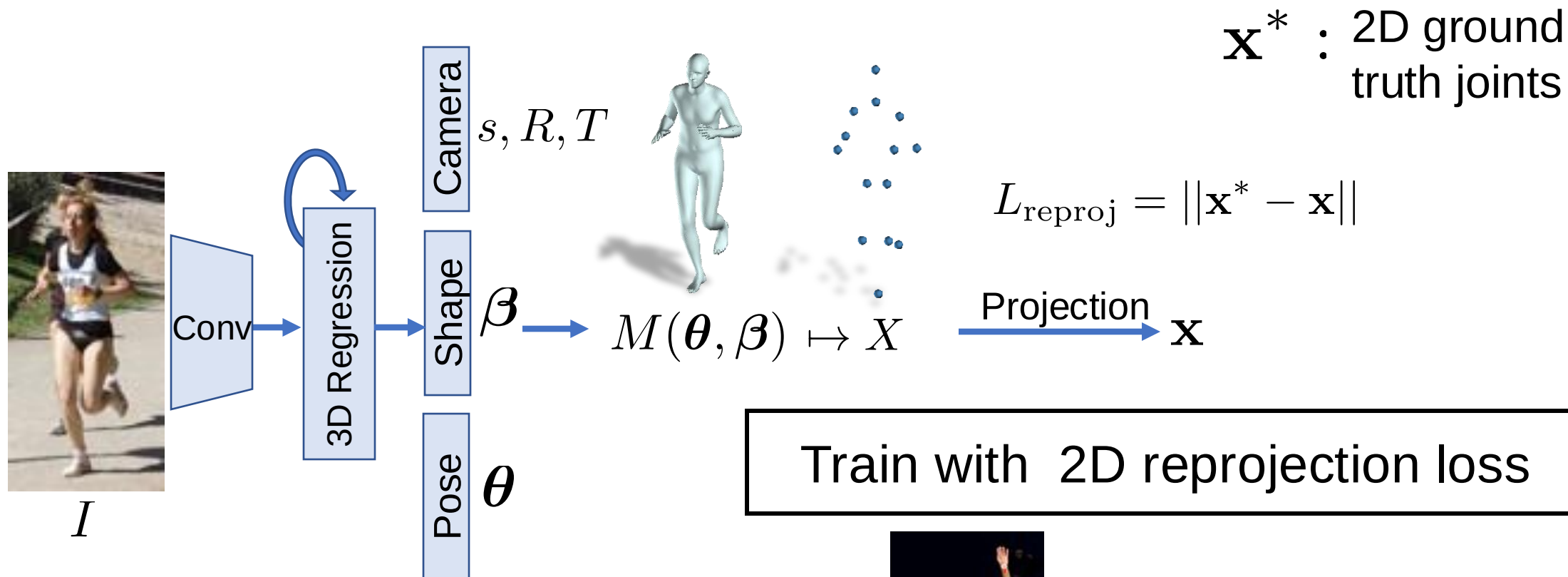
Jathushan Rajasegaran

Angjoo Kanazawa

Jitendra Malik

UC Berkeley

Human Mesh Recovery (HMR) 1.0



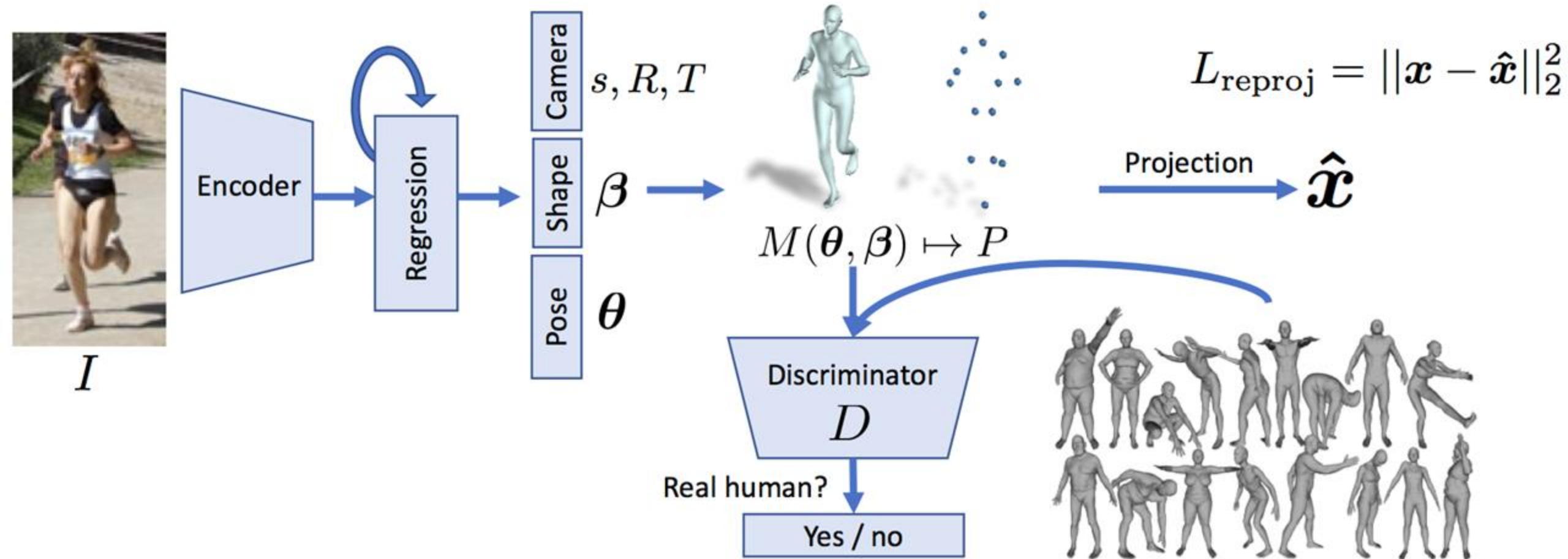
Kanazawa, Black, Jacobs and Malik,
CVPR 2018

$$\left\| \text{Image} - \Pi(\text{Mesh}) \right\|_2^2$$

Without any 2D-to-3D supervision...



Approach (2018)



Approach (2023)

New Data:
AVA (60hr) +
InstaVariety (200hr) +
AI Challenger (200k)



I

Encoder

Resnet50

ViT
(Pretrained)

MAE ImageNet
2D-KeyPt COCO+...

Regression

Camera

s, R, T

Shape

β

Pose

θ

$M(\theta, \beta) \mapsto P$

Discriminator
 D

Real human?

Yes / no

Pseudo-GT 3D Fitting
(ProHMR)

ViTPose

OpenPose

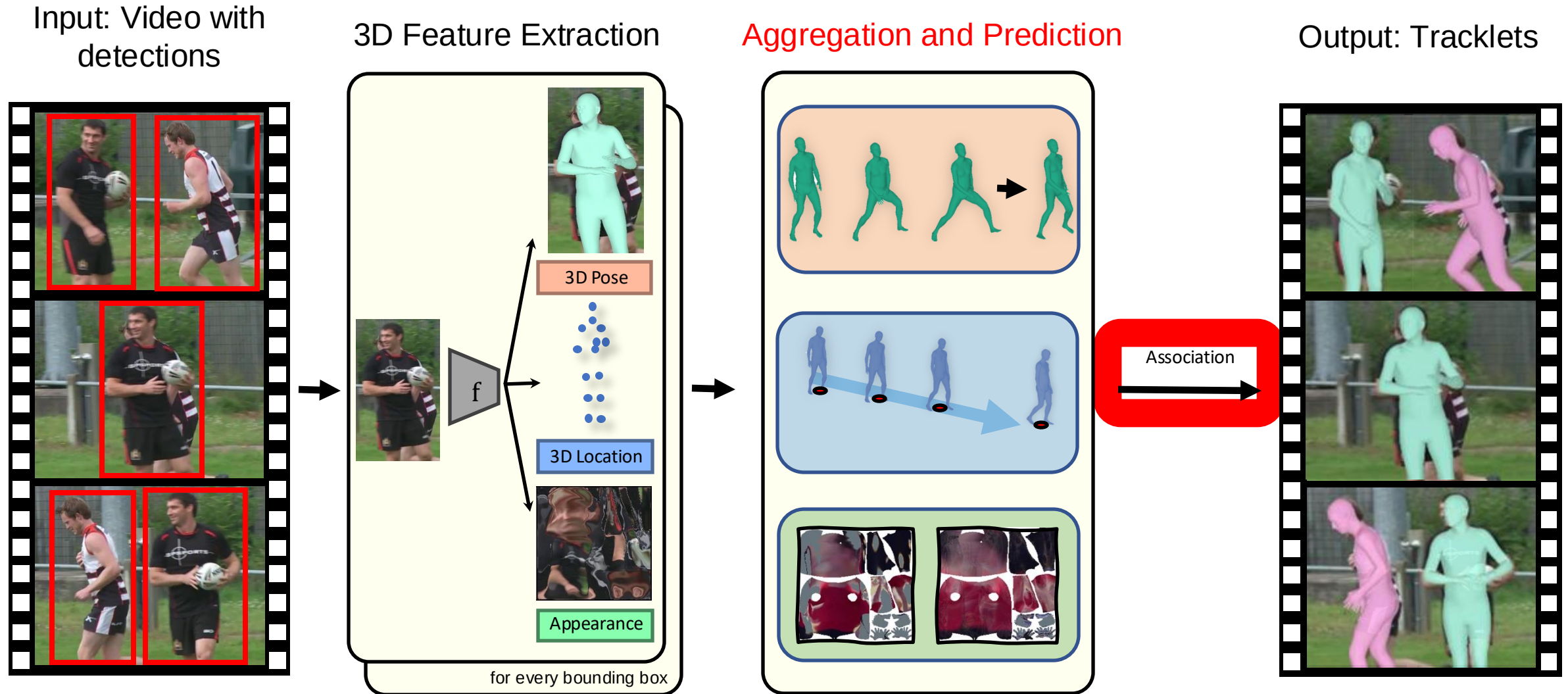
$$L_{\text{reproj}} = ||x - \hat{x}||_2^2$$

Projection

$\hat{x}$



PHALP: Tracking Humans by Predicting 3D Appearance, 3D Location & 3D Pose (CVPR 2022)



Insight #1

- In 2D, occlusion is hard to disambiguate



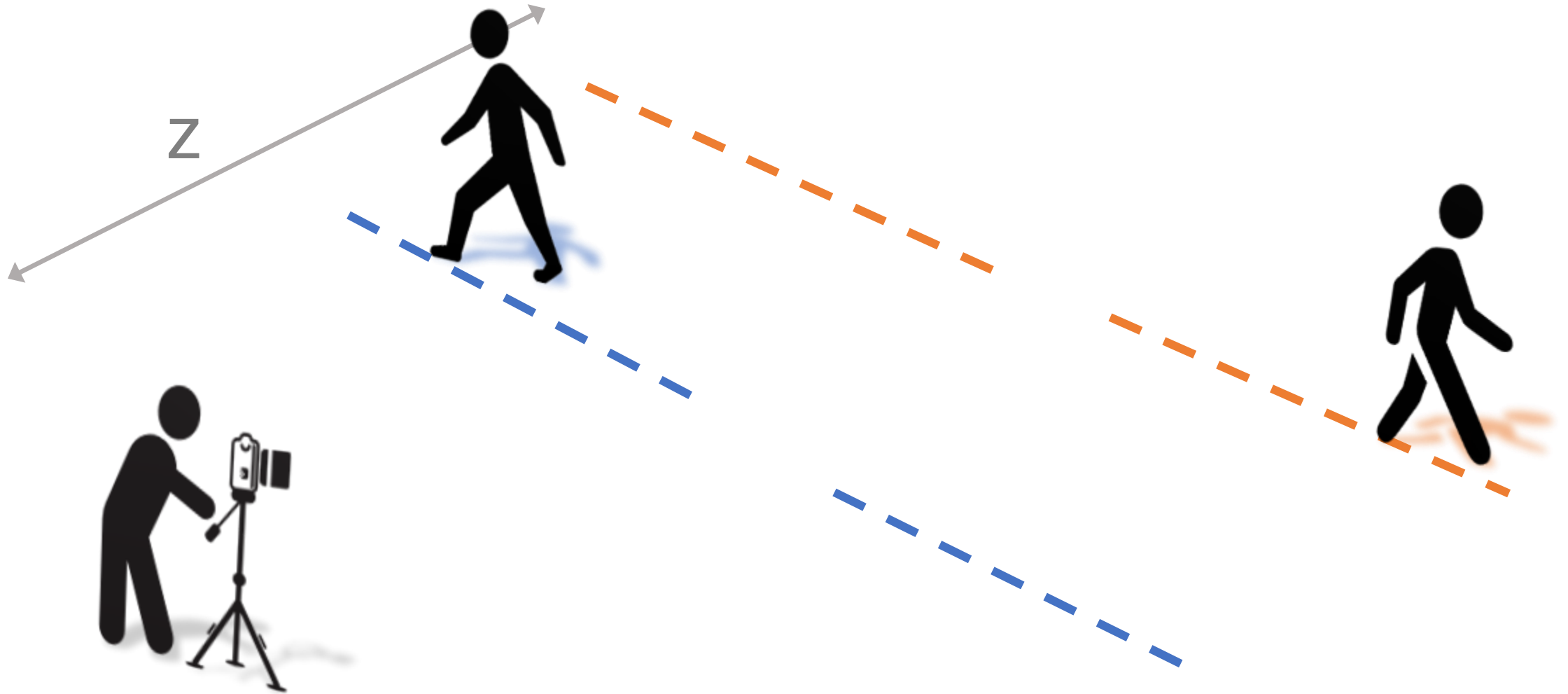
Insight #1

- In 2D, occlusion is hard to disambiguate



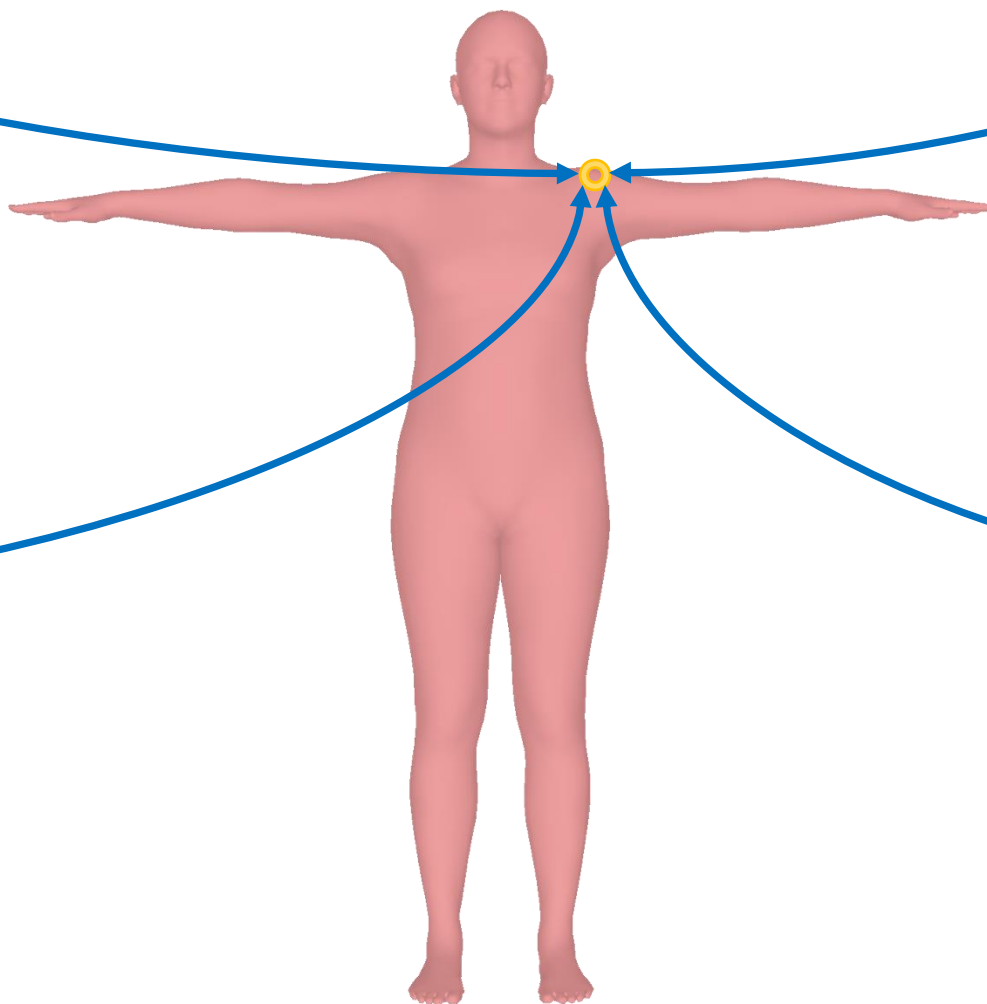
Insight #1

- In 2D, they overlap, but in 3D they don't!



Insight #2

3D appearance is invariant to
viewpoint and pose







What are the next steps?

Humanoid Locomotion as Next Token Prediction

Ilija Radosavovic Bike Zhang Baifeng Shi Jathushan Rajasegaran Sarthak Kamat

Trevor Darrell Koushil Sreenath Jitendra Malik

University of California, Berkeley





Humanoid Control as Next Token Prediction

Data

Neural network policy

Model based controller

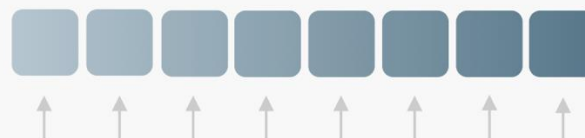


Motion capture

Internet videos



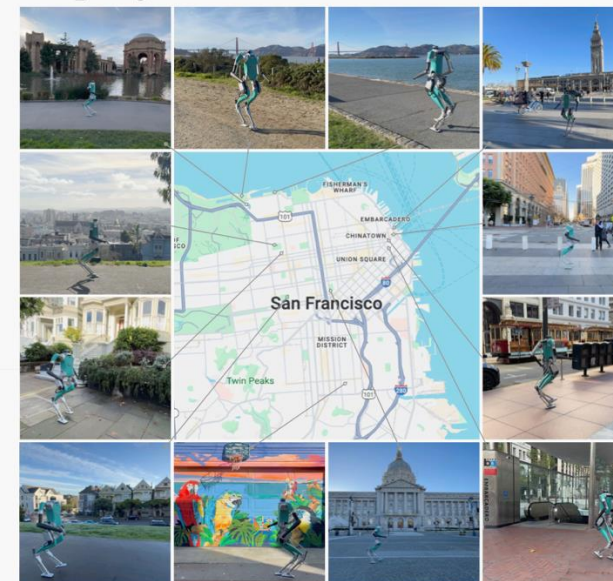
Training



Transformer

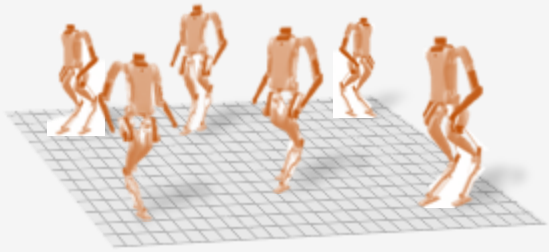


Deployment

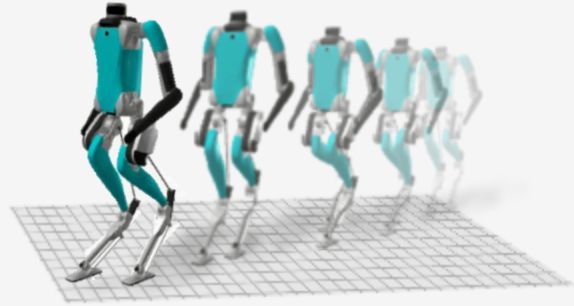


A Dataset of Walking Trajectories

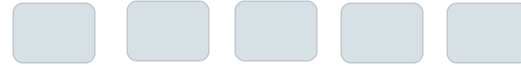
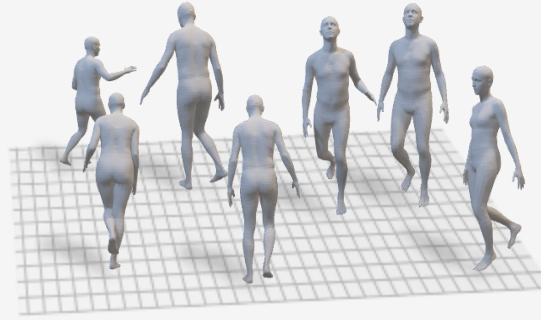
Neural Net Controller



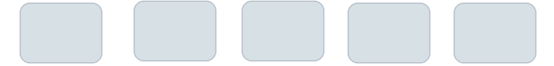
Model based Controller



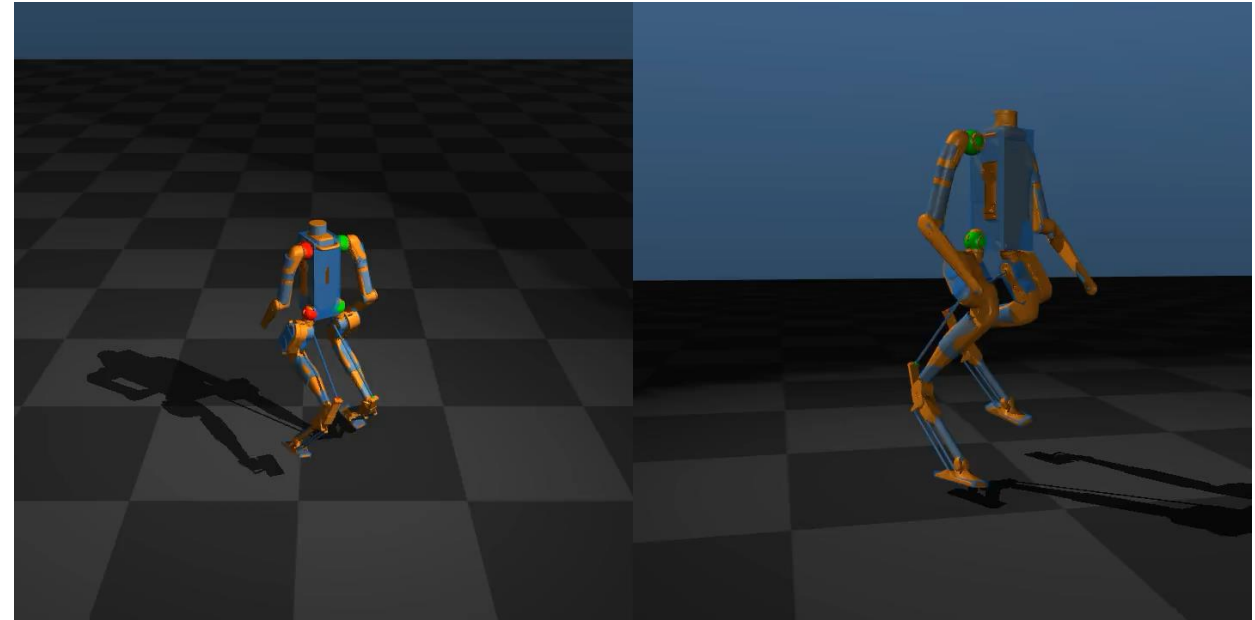
MoCap



Internet Videos

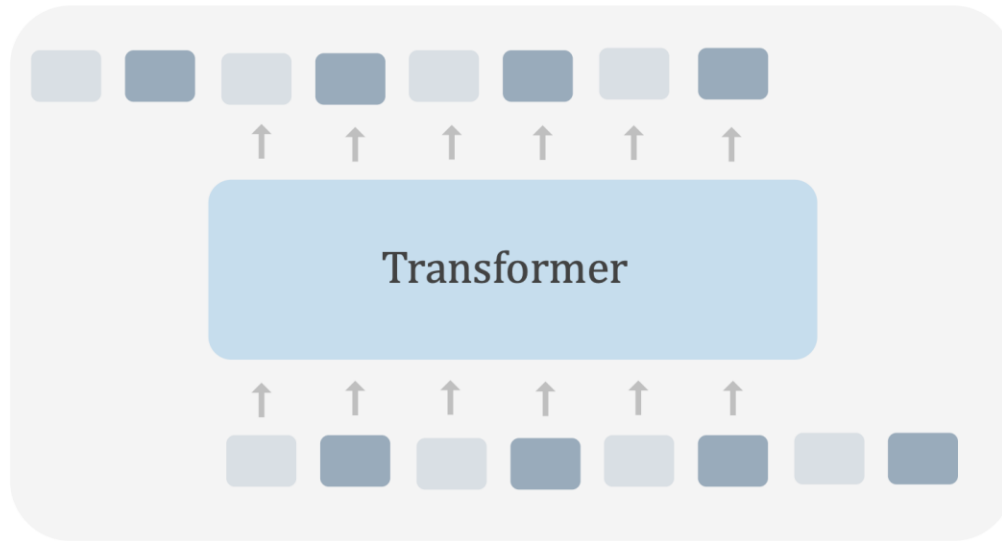


Retarget YouTube Video in Simulation

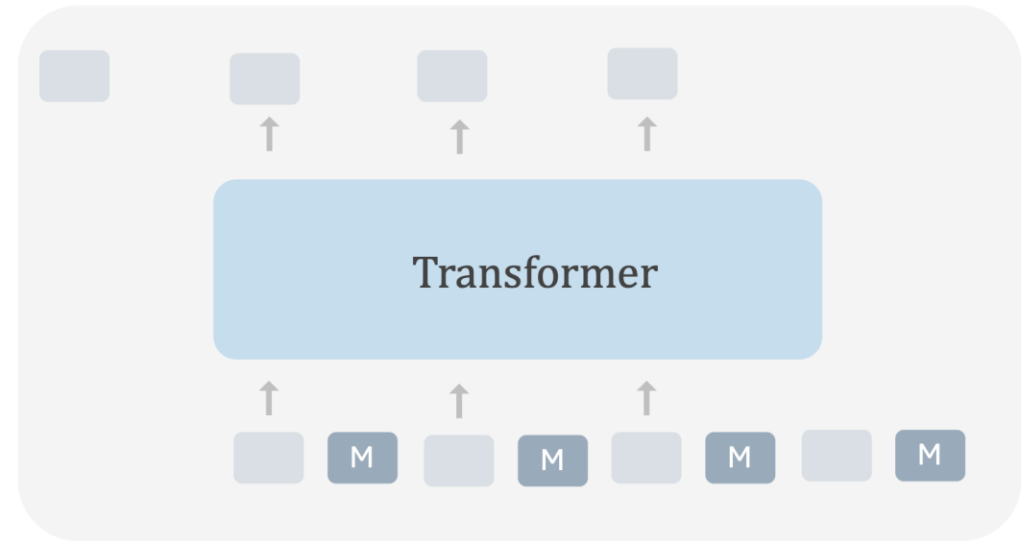


Training with Missing Data

Training with complete data



Training with missing data



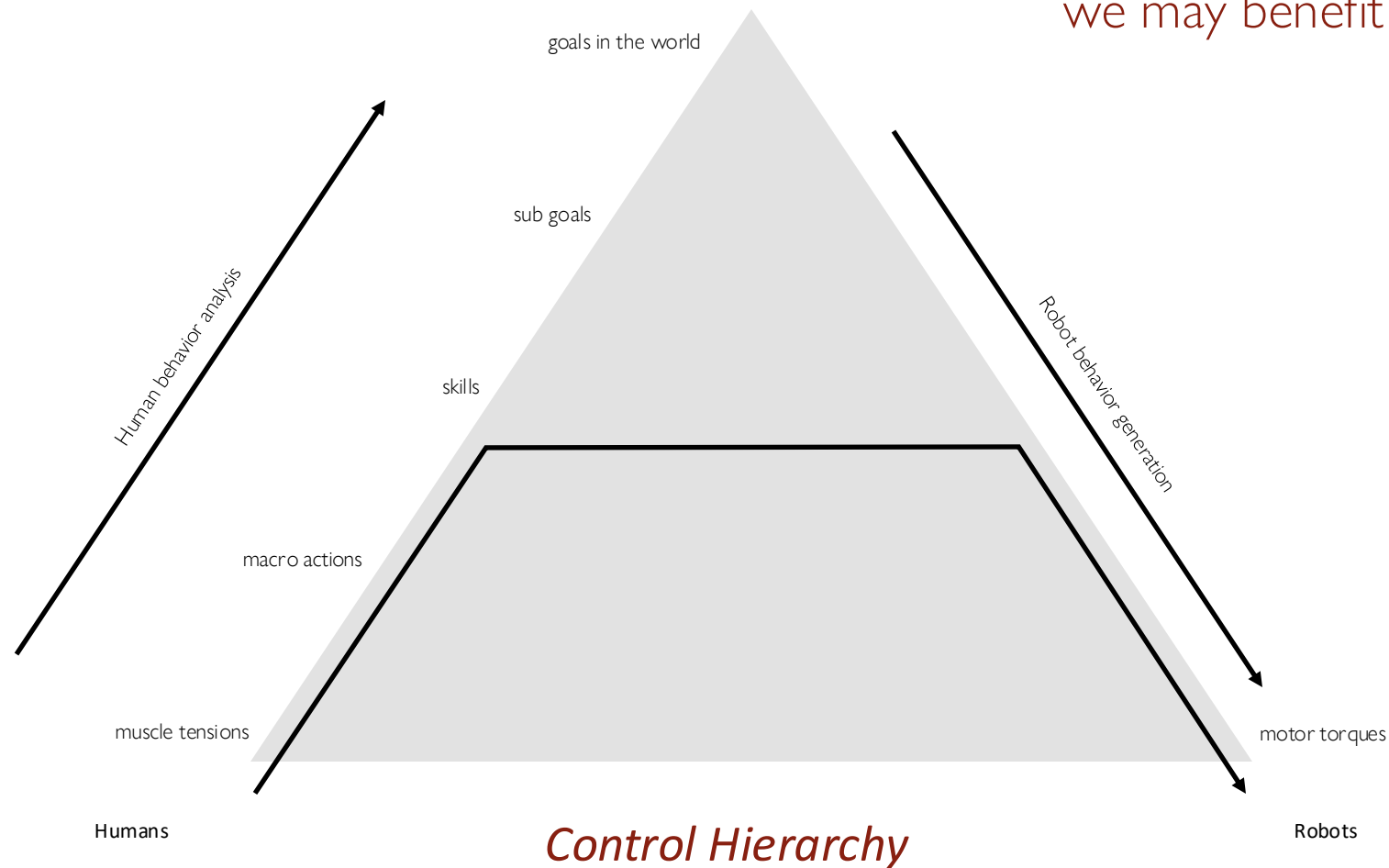
Learning at different abstraction levels

Credit: Saurabh Gupta, UIUC

Depending on the amount of gap between:

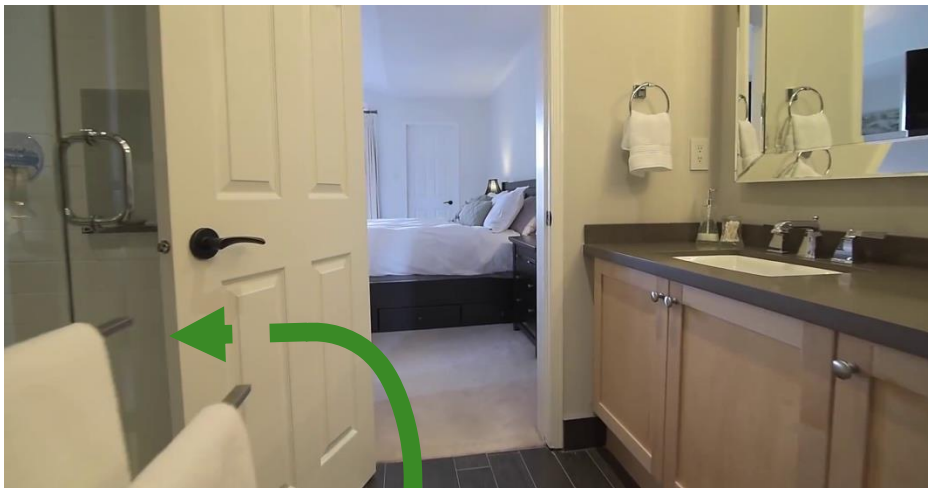
- goals,
- embodiment,
- what we can observe in videos

we may benefit from transfer at different levels.

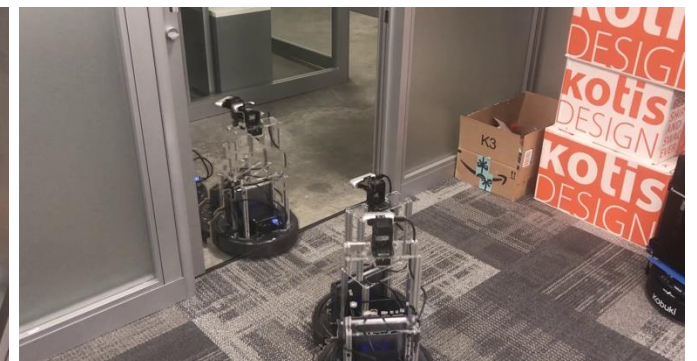


We have found success with applying this idea to navigation

- Affordances (What can you do here)
- Subroutines (How)



Turn left



Learning Navigation by Watching Videos

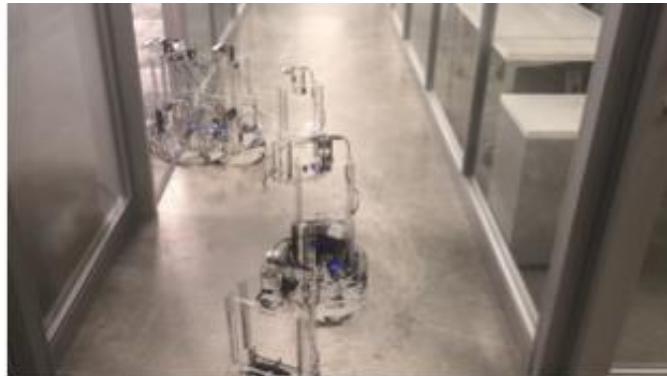


**Robot Interacting
with Environments**

+



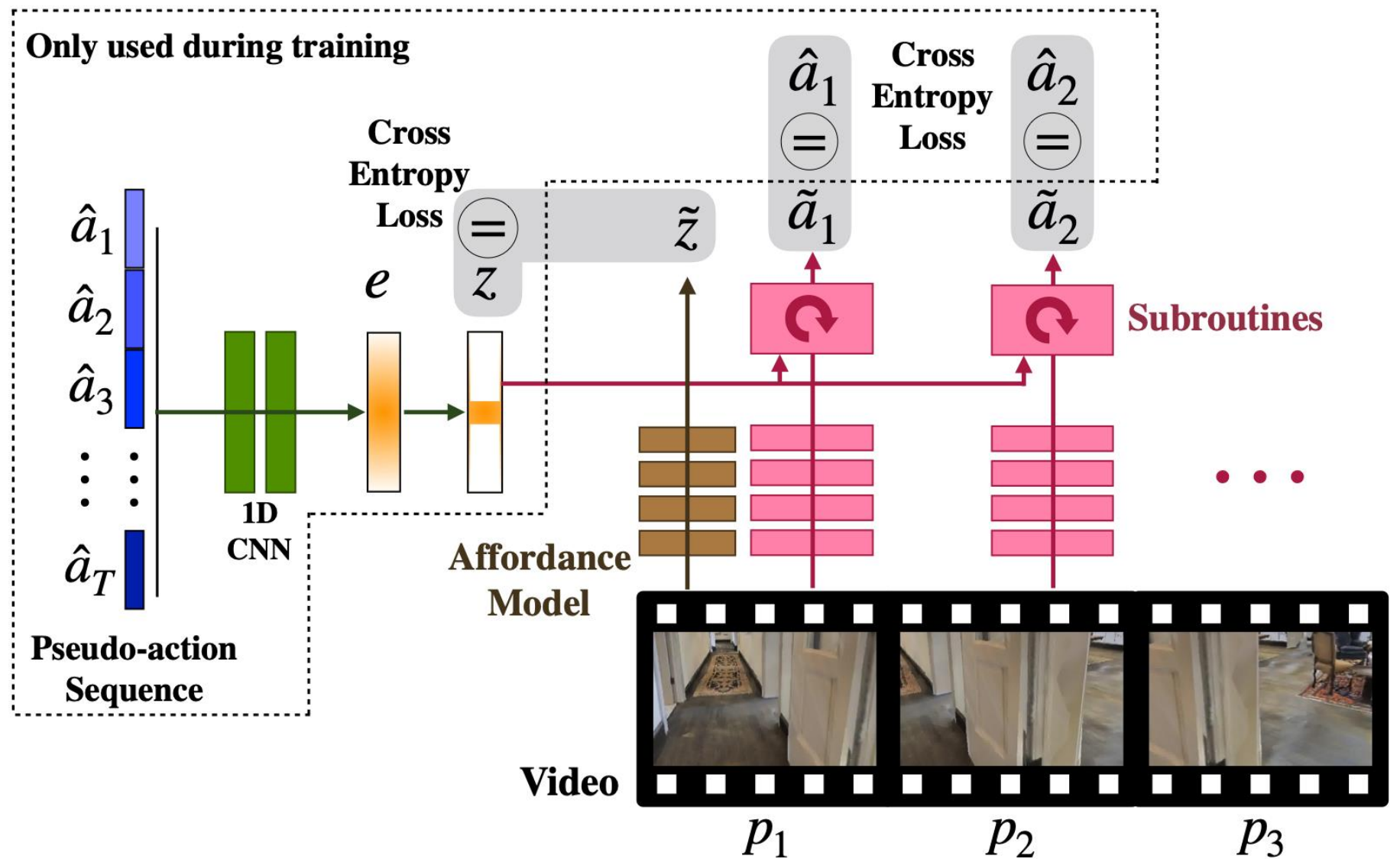
**First-person Videos
from Walking People**



**Visuomotor Subroutines:
Short-horizon policies with
consistent and diverse behavior.**



**Affordance Model:
Predict what subroutine
to use where.**



SubR1 - Turns Right

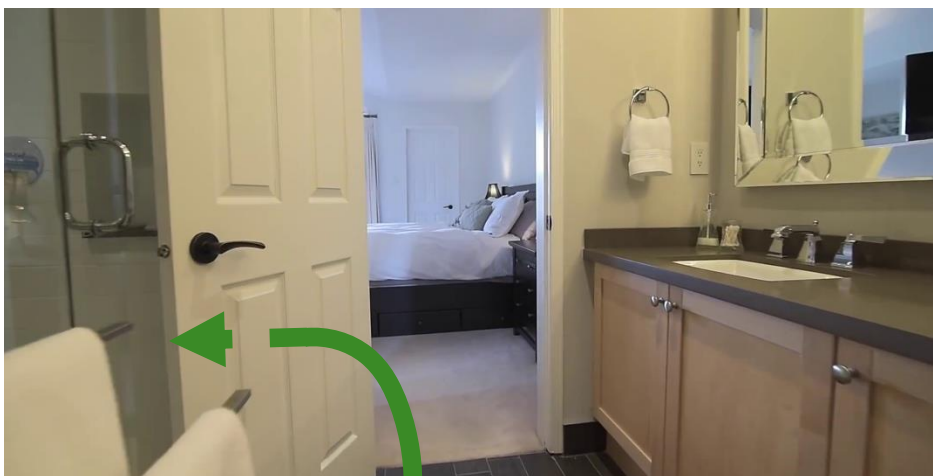


SubR3 - Turns Left



We have found success with applying this idea to navigation

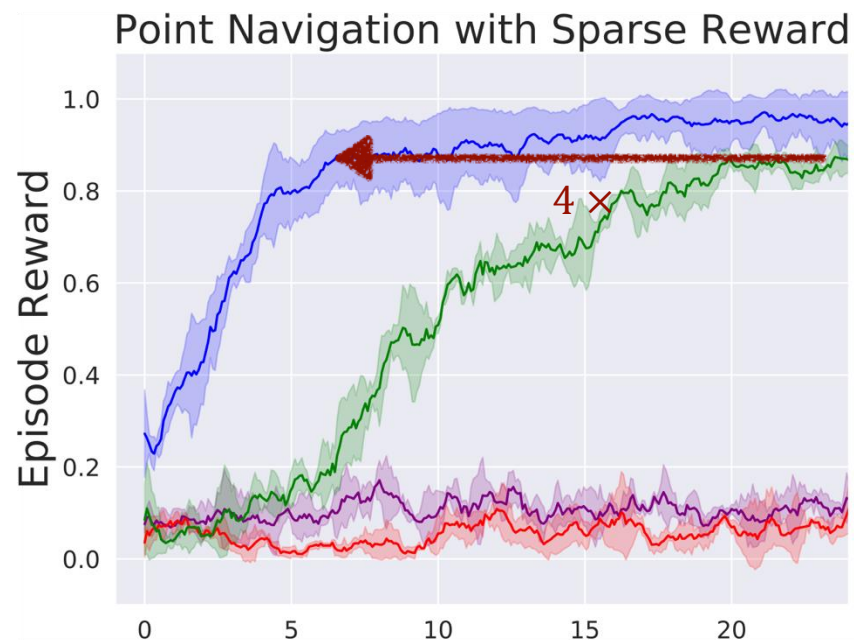
- Affordances (What can you do here)
- Subroutines (How)



a) Use subroutines and affordances for exploration



b) Use subroutines and affordances for hierarchical RL



INSTRUCTIONS ON HOW TO CLIMB A STAIRCASE

No one will have failed to observe that frequently the floor bends in such a way that one part rises at a right angle to the plane formed by the floor and then the following section arranges itself parallel to the flatness, so as to provide a step to a new perpendicular, a process which is repeated in a spiral or in a broken line to highly variable elevations. Ducking down and placing the left hand on one of the vertical parts and the right hand upon the corresponding horizontal, one is in momentary possession of a step or stair. Each one of these steps, formed as we have seen by two elements, is situated somewhat higher and further than the one prior, a principle which gives the idea of a staircase, while whatever other combination, producing perhaps more beautiful or picturesque shapes, would be incapable of translating one from the ground floor to the first floor.

You tackle a stairway face on, for if you try it backwards or sideways, it ends up being particularly uncomfortable. The natural stance consists of holding oneself upright, arms hanging easily at the sides, head erect but not so

much so that the eyes no longer see the steps immediately above, while one tramps up, breathing lightly and with regularity. To climb a staircase one begins by lifting that part of the body located below and to the right, usually encased in leather or deerskin, and which, with a few exceptions, fits exactly on the stair. Said part set down on the first step (to abbreviate we shall call it "the foot"), one draws up the equivalent part on the left side (also called "foot" but not to be confused with "the foot" cited above), and lifting this other part to the level of "the foot," makes it continue along until it is set in place on the second step, at which point the foot will rest, and "the foot" will rest on the first. (The first steps are always the most difficult, until you acquire the necessary coordination. The coincidence of names between the foot and "the foot" makes the explanation more difficult. Be especially careful not to raise, at the same time, the foot and "the foot.")

Having arrived by this method at the second step, it's easy enough to repeat the movements alternately, until one reaches the top of the staircase. One gets off it easily, with a light tap of the heel to fix it in place, to make sure it will not move until one is ready to come down.

Julio Cortazar



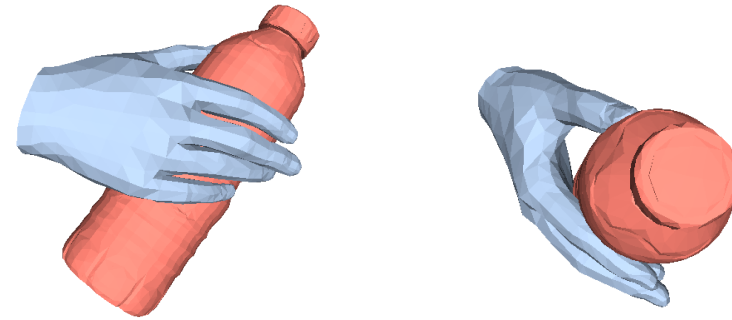
Lecture 8B: Visual Imitation

Jitendra Malik

Imitation Learning for Robot Manipulation



image



3D reconstruction

Reconstructing Hand-Object Interactions in the Wild

Zhe Cao, Ilija Radosavovic, Angjoo Kanazawa, Jitendra Malik (ICCV 2021)



Reconstructing Hands in 3D with Transformers

Georgios Pavlakos, Dandan Shan, Ilija Radosavovic, Angjoo Kanazawa, David Fouhey, Jitendra Malik

HaMeR Results



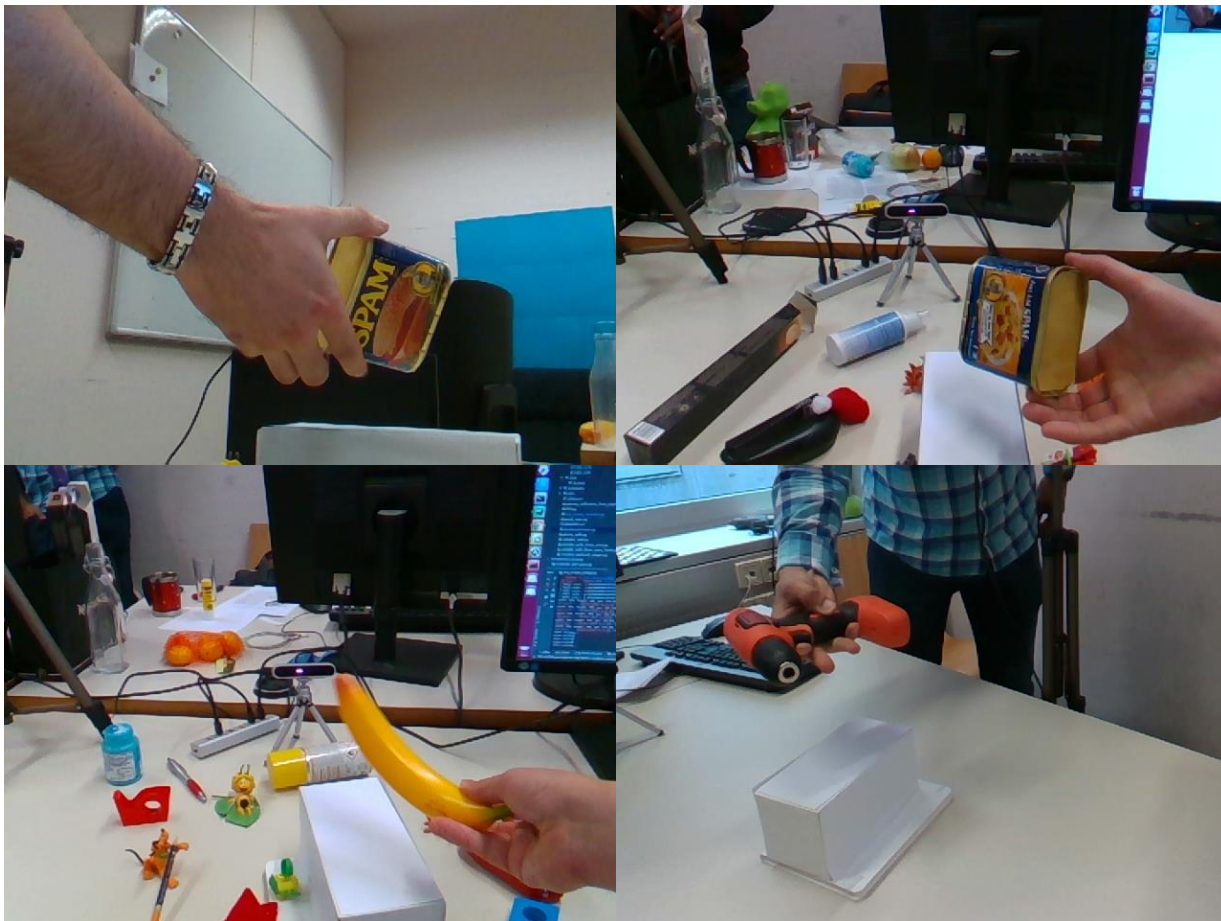
Evaluation - FreiHAND



3D Pose / 3D Mesh Accuracy

Method	PA-MPJPE ↓	PA-MPVPE ↓	F@5 ↑	F@15 ↑
I2L-MeshNet [39]	7.4	7.6	0.681	0.973
Pose2Mesh [10]	7.7	7.8	0.674	0.969
I2UV-HandNet [7]	6.7	6.9	0.707	0.977
METRO [33]	6.5	6.3	0.731	0.984
Tang <i>et al.</i> [53]	6.7	6.7	0.724	0.981
Mesh Graphormer [34]	5.9	6.0	0.764	0.986
MobRecon [8]	5.7	5.8	0.784	0.986
AMVUR [24]	6.2	6.1	0.767	0.987
Ours	6.0	5.7	0.785	0.990

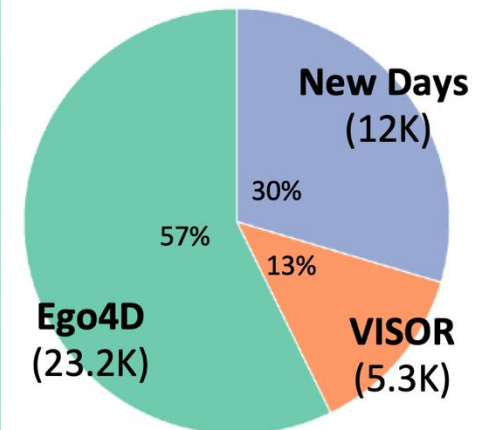
Evaluation - HO3D



3D Pose / 3D Mesh Accuracy

Method	AUC _J ↑	PA-MPJPE ↓	AUC _V ↑	PA-MPVPE ↓	F@5 ↑	F@15 ↑
Liu <i>et al.</i> [36]	0.803	9.9	0.810	9.5	0.528	0.956
HandOccNet [44]	0.819	9.1	0.819	8.8	0.564	0.963
I2UV-HandNet [7]	0.804	9.9	0.799	10.1	0.500	0.943
Hampali <i>et al.</i> [19]	0.788	10.7	0.790	10.6	0.506	0.942
Hasson <i>et al.</i> [21]	0.780	11.0	0.777	11.2	0.464	0.939
ArtiBoost [58]	0.773	11.4	0.782	10.9	0.488	0.944
Pose2Mesh [10]	0.754	12.5	0.749	12.7	0.441	0.909
I2L-MeshNet [39]	0.775	11.2	0.722	13.9	0.409	0.932
METRO [33]	0.792	10.4	0.779	11.1	0.484	0.946
MobRecon[8]	-	9.2	-	9.4	0.538	0.957
Keypoint Trans [20]	0.786	10.8	-	-	-	-
AMVUR [24]	0.835	8.3	0.836	8.2	0.608	0.965
Ours	0.846	7.7	0.841	7.9	0.635	0.980

Hint - Hand Interactions Dataset



New Days of Hands



Epic Kitchens



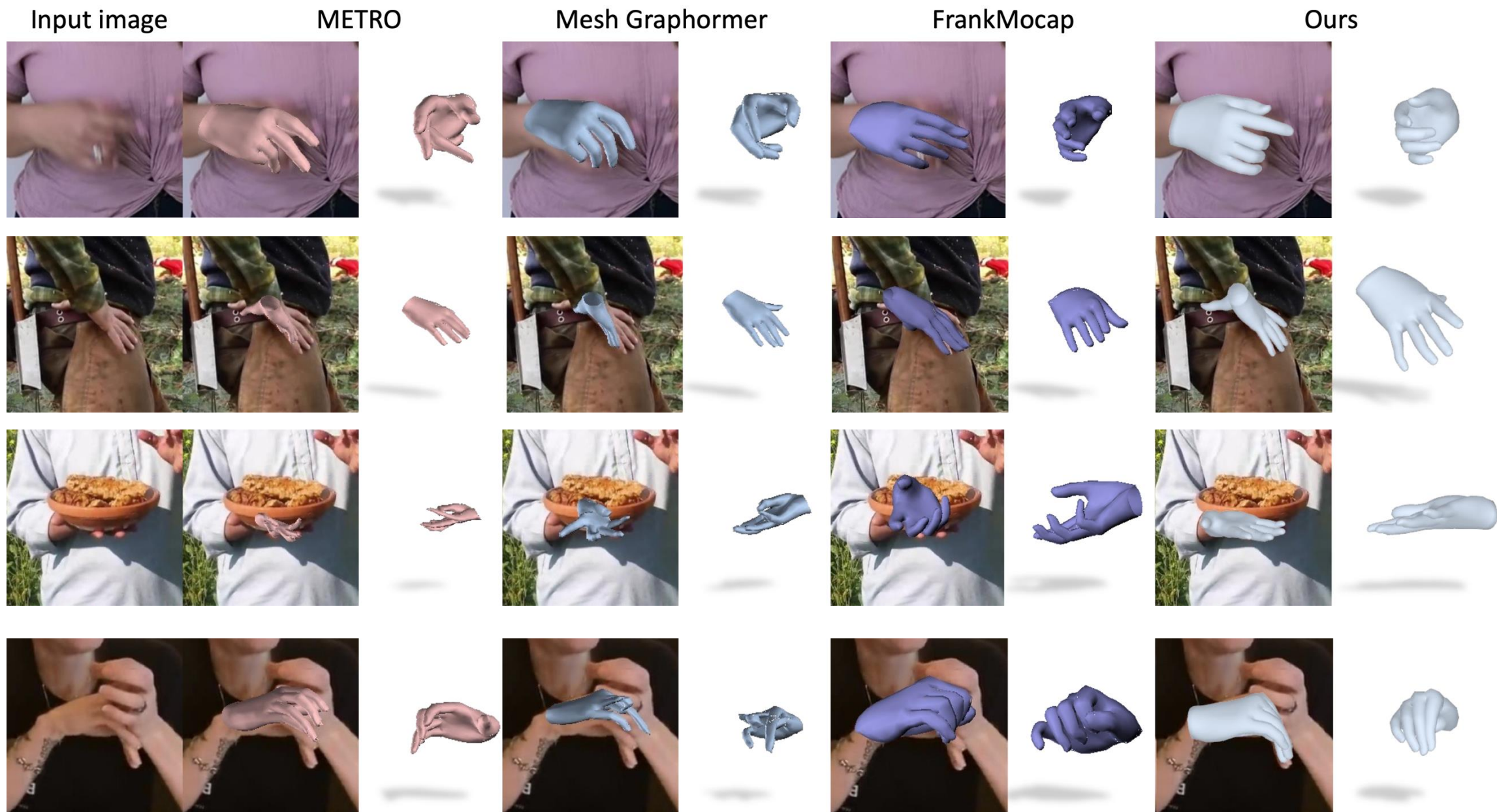
Ego4D



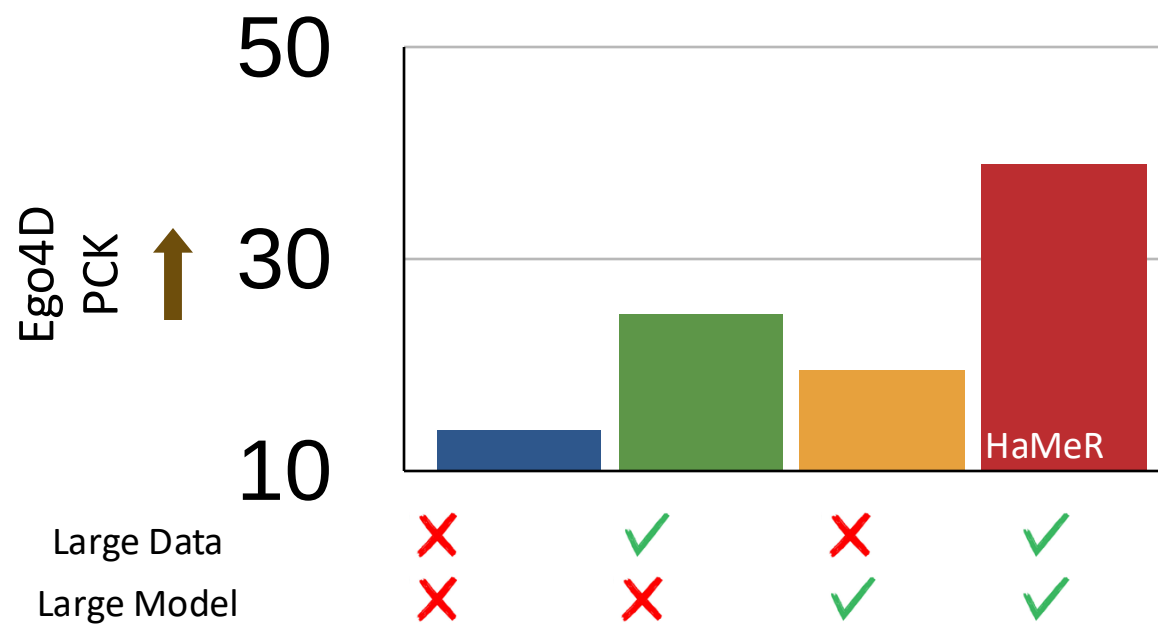
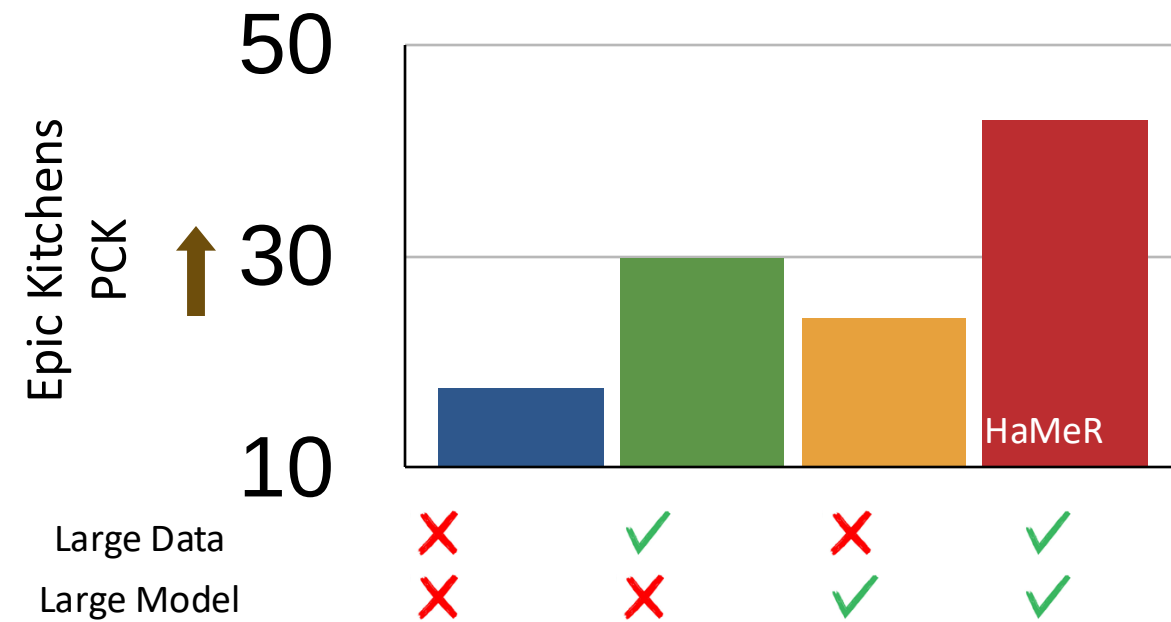
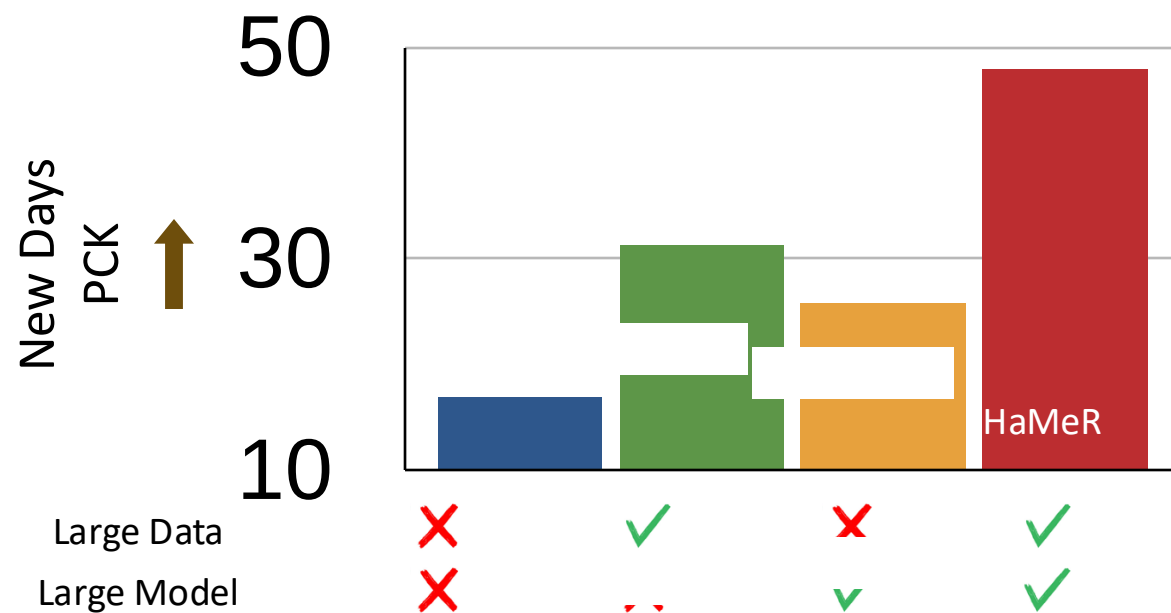
Evaluation - Hint

	Method	New Days		
		@0.05	@0.1	@0.15
All Joints	FrankMocap [49]	16.1	41.4	60.2
	METRO [33]	14.7	38.8	57.3
	MeshGraphormer [34]	16.8	42.0	59.7
	HandOccNet (param) [44]	9.1	28.4	47.8
	HandOccNet (no param) [44]	13.7	39.1	59.3
	Ours	48.0	78.0	88.8

Comparisons with the State of the Art



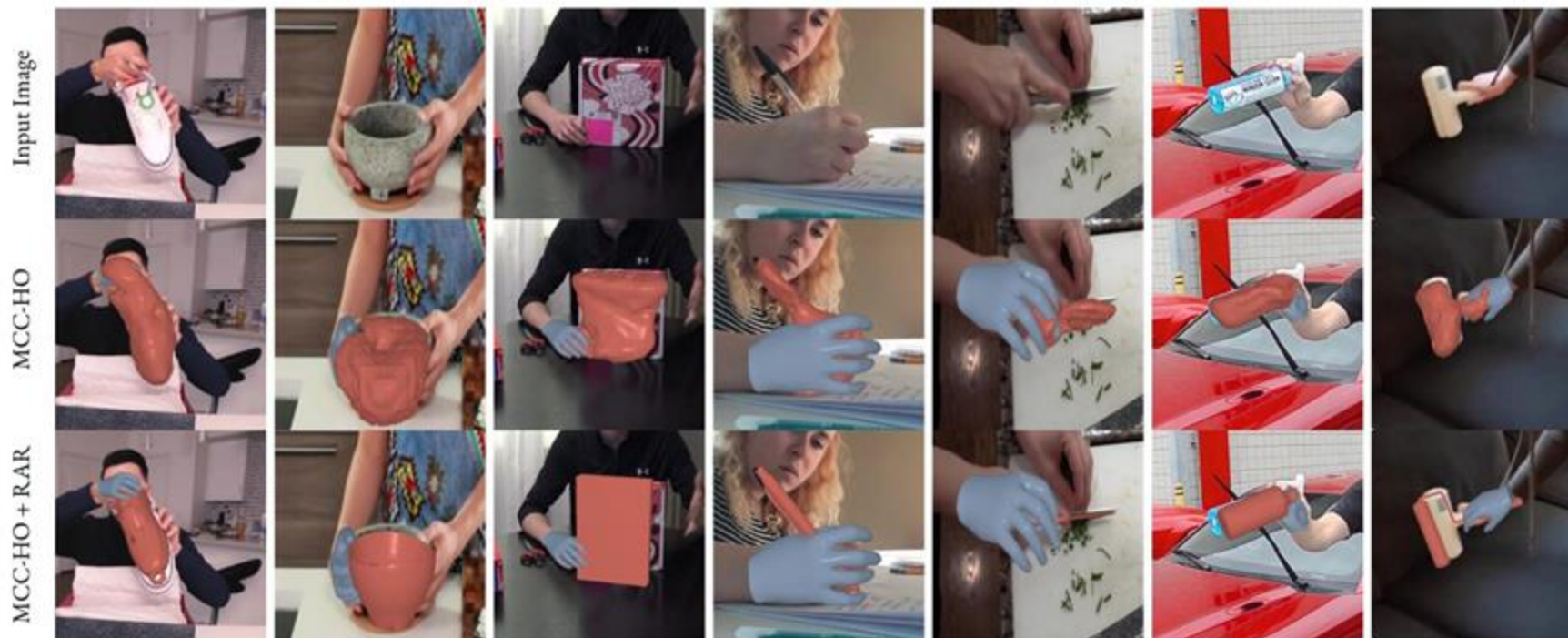
The effect of Data and Model



Reconstructing Hand-Held Objects in 3D

Jane Wu, Georgios Pavlakos, Georgia Gkioxari, Jitendra Malik

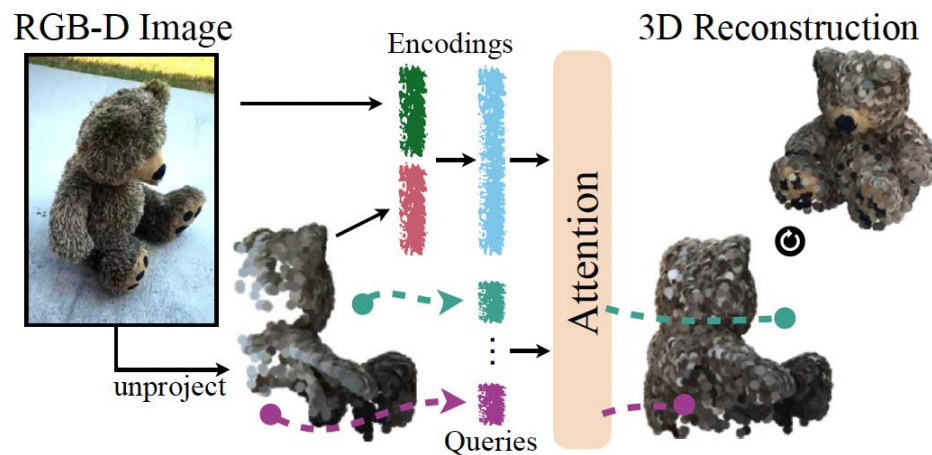
<https://arxiv.org/abs/2404.06507>



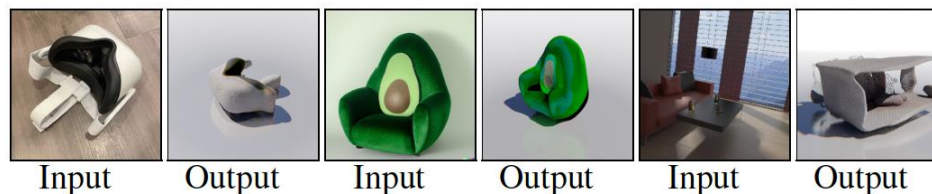
Builds on two previous foundation models : HaMeR (Berkeley) and MCC (Meta)

Multiview Compressive Coding for 3D Reconstruction

CY Wu, J Johnson, J Malik, C Feichtenhofer, G. Gkioxari

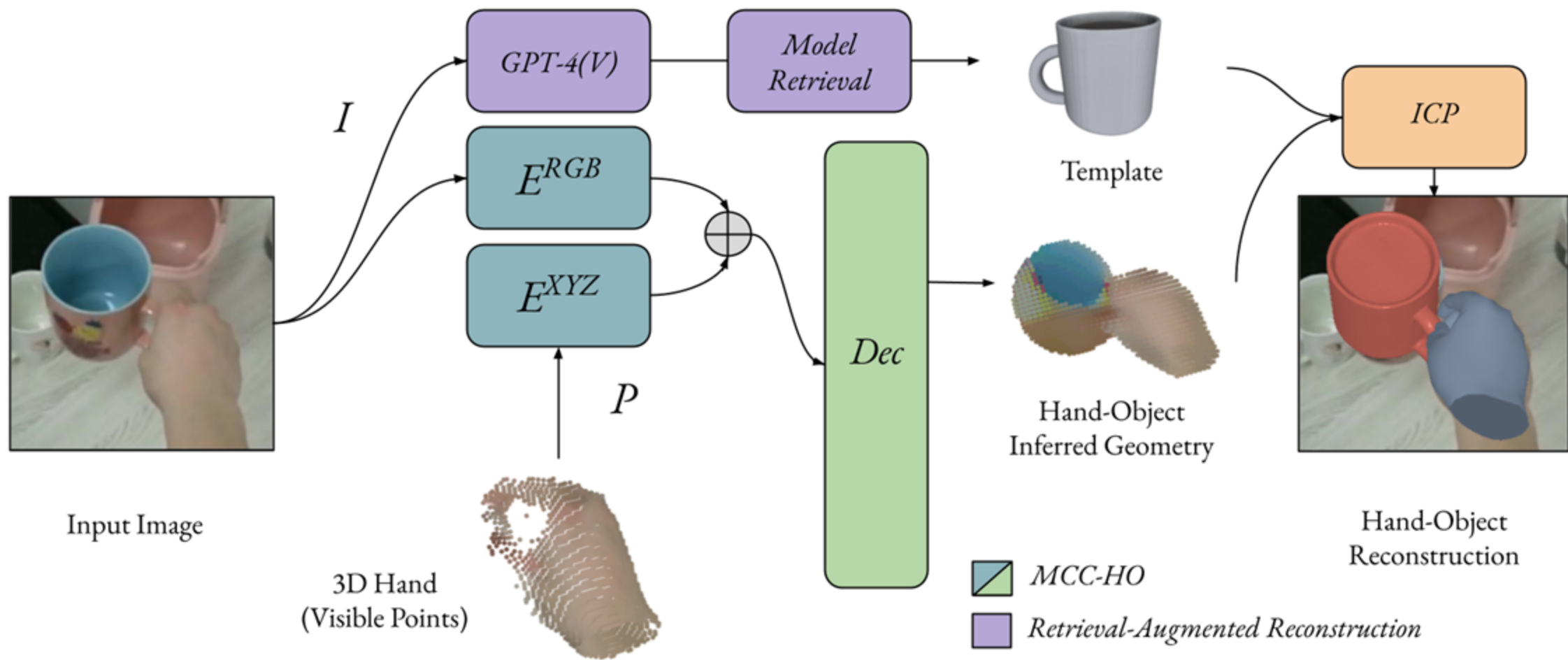


(a) MCC Overview



(b) 3D Reconstructions by MCC

Figure 1. **Multiview Compressive Coding (MCC)**. (a): MCC encodes an input RGB-D image and uses an attention-based model to predict the occupancy and color of query points to form the final 3D reconstruction. (b): MCC generalizes to novel objects captured with iPhones (left) or imagined by DALL·E 2 [47] (middle). It is also general – it works not only on objects but also scenes (right).



Reconstruction

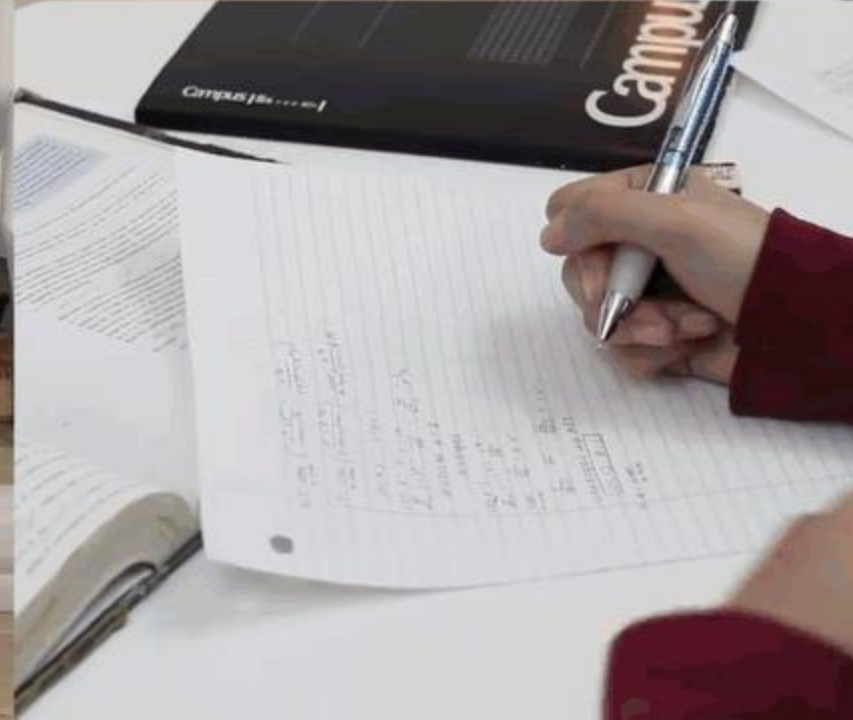
MCC-HO
Results



DexYCB				MOW			HOI4D		
	F-5 (↑)	F-10 (↑)	CD (↓)	F-5 (↑)	F-10 (↑)	CD (↓)	F-5 (↑)	F-10 (↑)	CD (↓)
HO [24]	0.24	0.48	4.76	0.03	0.06	49.8	0.28	0.51	3.86
IHOI [90]	-	-	-	0.13	0.24	23.1	0.42	0.70	2.7
MCC-HO	0.36	0.60	3.74	0.15	0.31	15.2	0.52	0.78	1.36

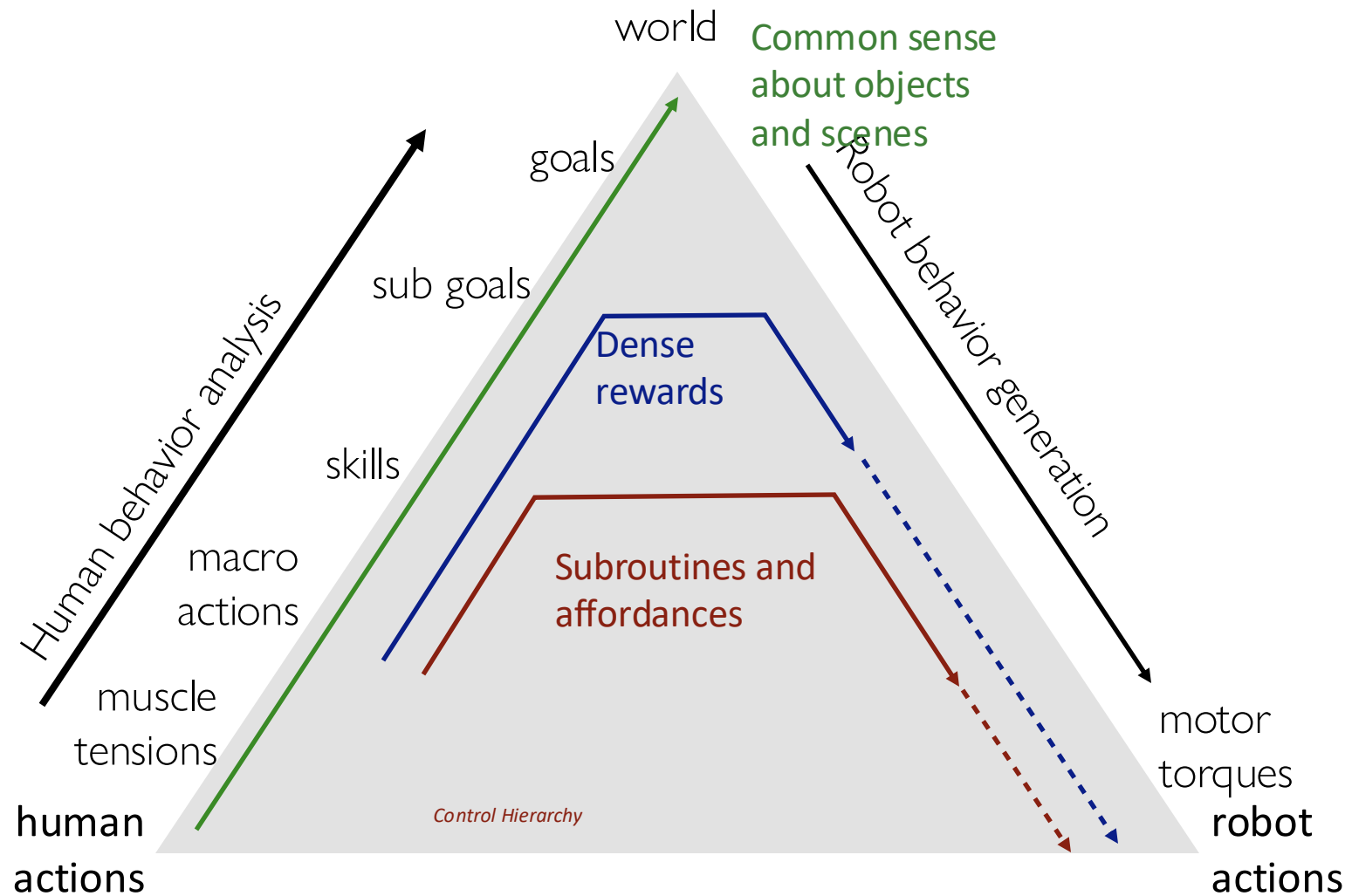
Comparison to prior work. Chamfer Distance (cm²) and F-score (5mm, 10mm).





Transferring at different abstraction levels

Credit: Saurabh Gupta, UIUC



Interactive Object Understanding

- A) Which sites can we interact at?
(cupboard handles)
- B) How to interact with those sites?
(using adducted thumb grasp)
- C) What happens when we do?
(the cupboard opens to reveal condiments)



*Learn through observation of **human hands** interacting with the world.*

Human Hands in Egocentric Videos are Informative



1. Attending to hands localizes and stabilizes active objects.
2. Hands show where all we can interact in the scene.
3. Analyzing hands reveals information about objects: their state and how to interact with them.

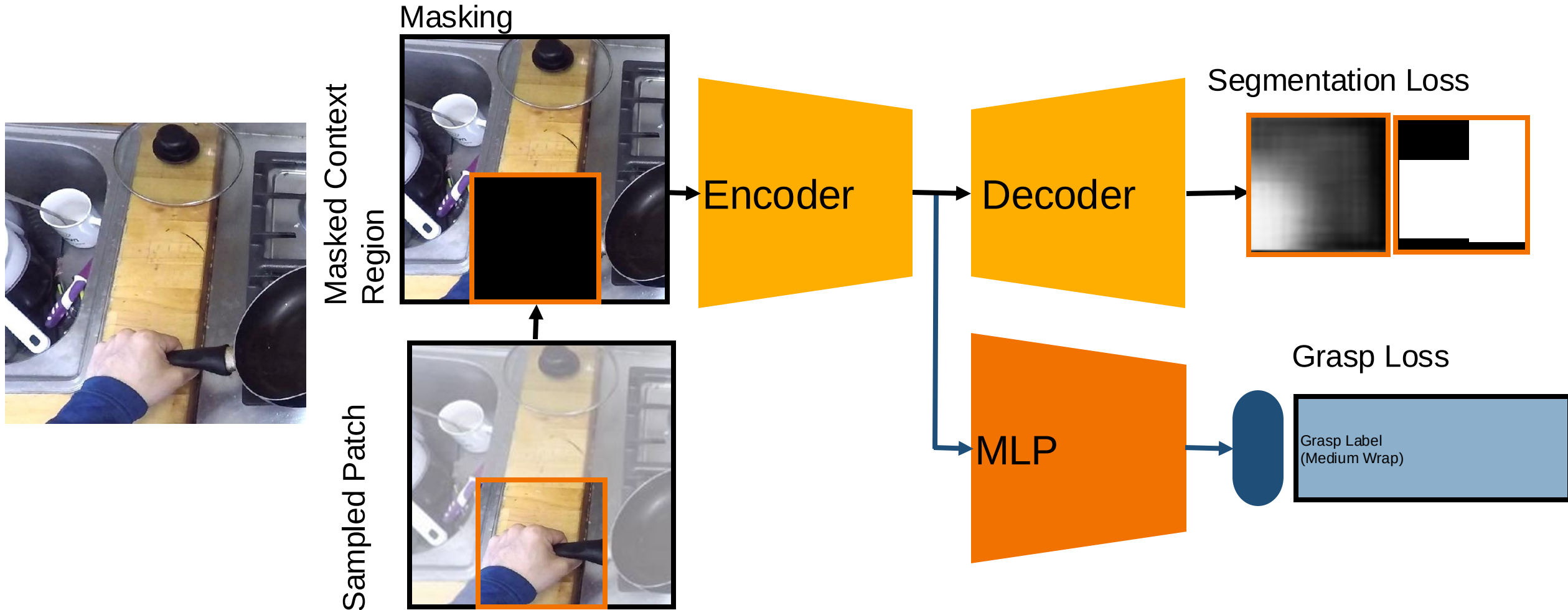
A,B) Learning Object Affordances

Data creation using off-the-shelf models

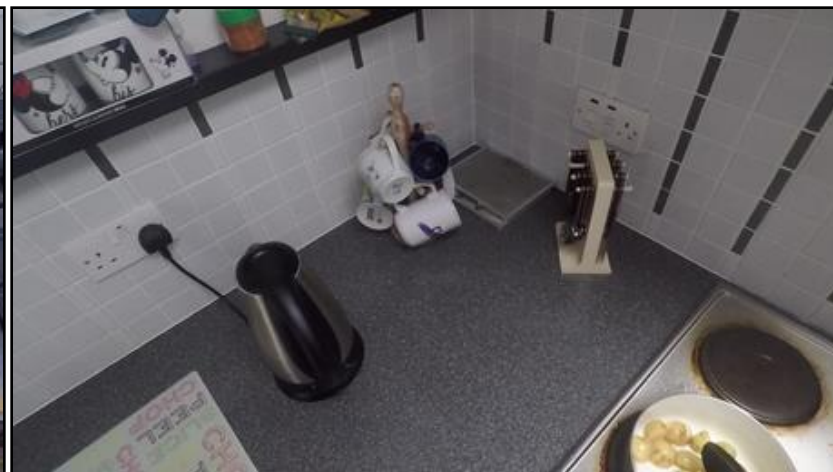


A,B) Learning Object Affordances

Affordance from Context Prediction (ACP)



Learning Object Affordances: Results



Learning Object Affordances: Results

Grasps Afforded by Objects (GAO) Task

Top Ranking Objects predicted by ACP



Precision Disk



Parallel Extension



Large Diameter



Power Sphere



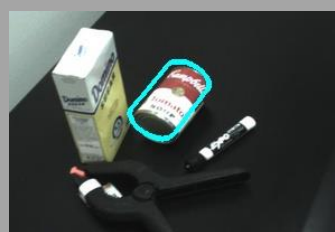
Sphere 4 Finger



Sphere 3 Finger



Medium Wrap



Interactive Object Understanding

A) Which sites can we interact at?
(cupboard handles)

B) How to interact with those sites?
(using adducted thumb grasp)

C) What happens when we do?
(the cupboard opens to reveal
condiments)



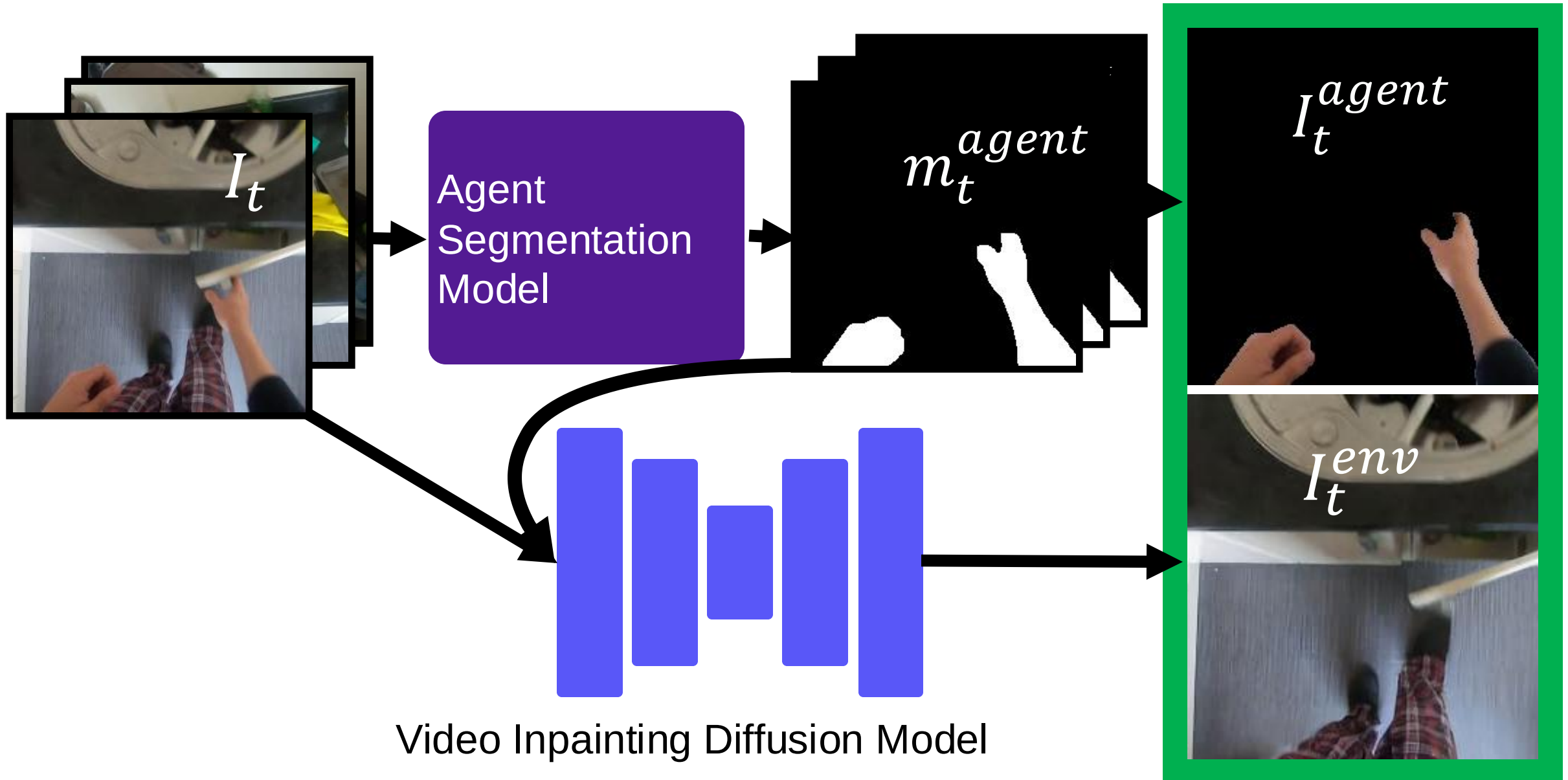
*Learn through observation of **human hands** interacting with the world.*

Look Ma, No Hands! Agent-Environment Factorization of Egocentric Videos

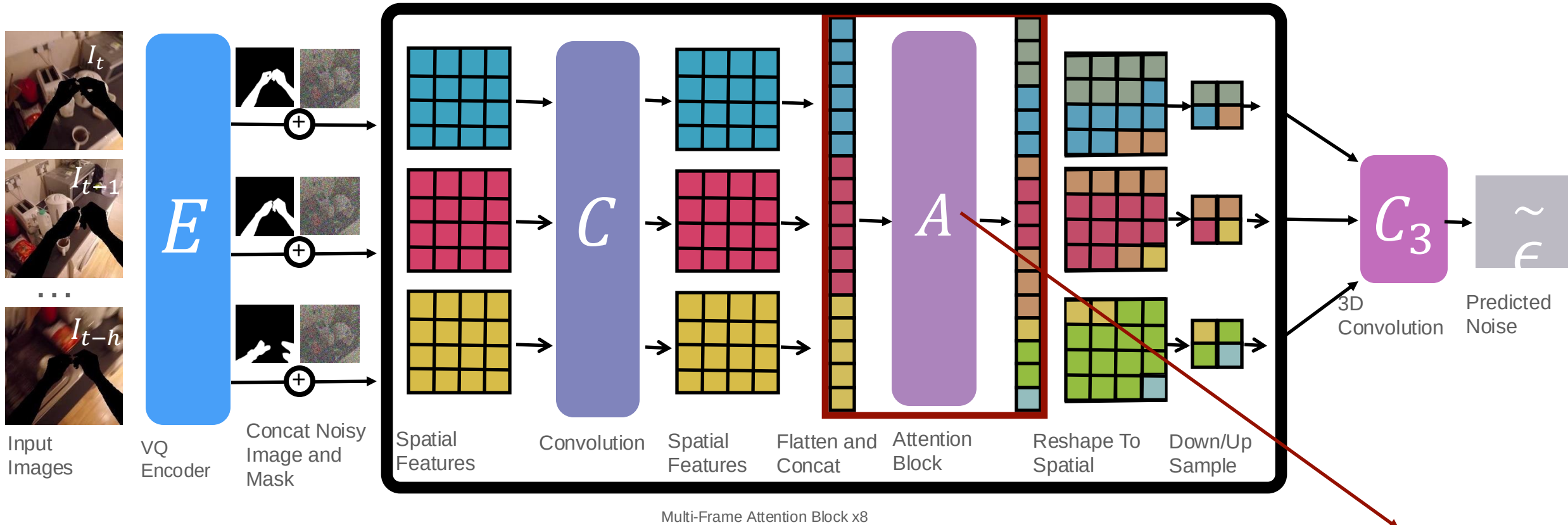
- Matthew Chang, Aditya Prakash, Saurabh Gupta



Agent-Environment *Factored* Representations



Video Inpainting Diffusion Model (VIDM)



Visualizations



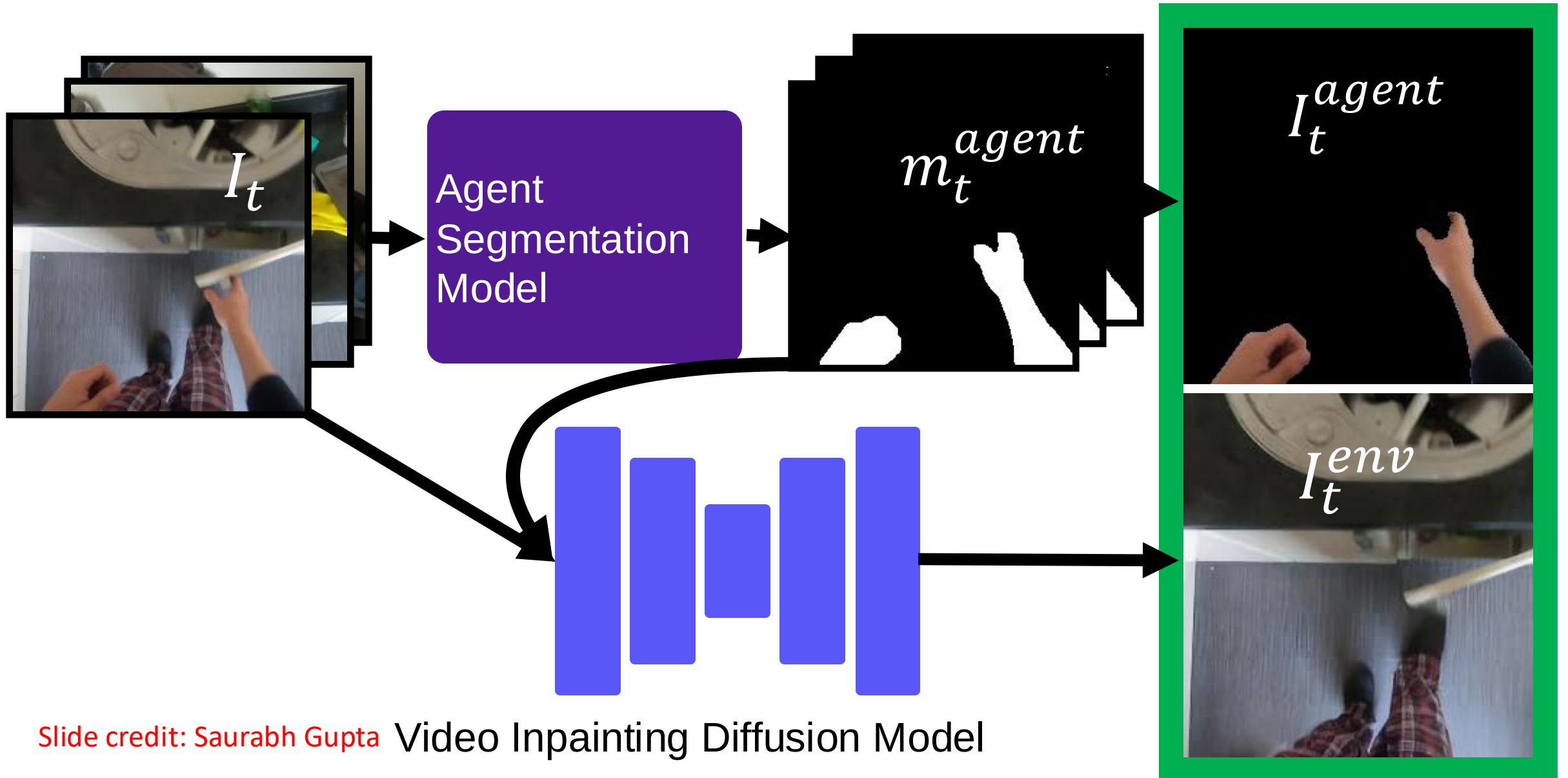
Visualizations



Visualizations



Agent-Environment *Factored* Representations



Using the *Factored* Representations

a) Affordances

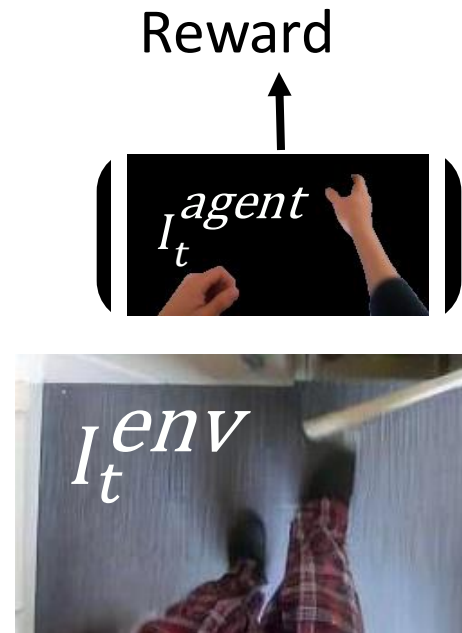


b) Reward functions from human data

Learn reward function
from human data



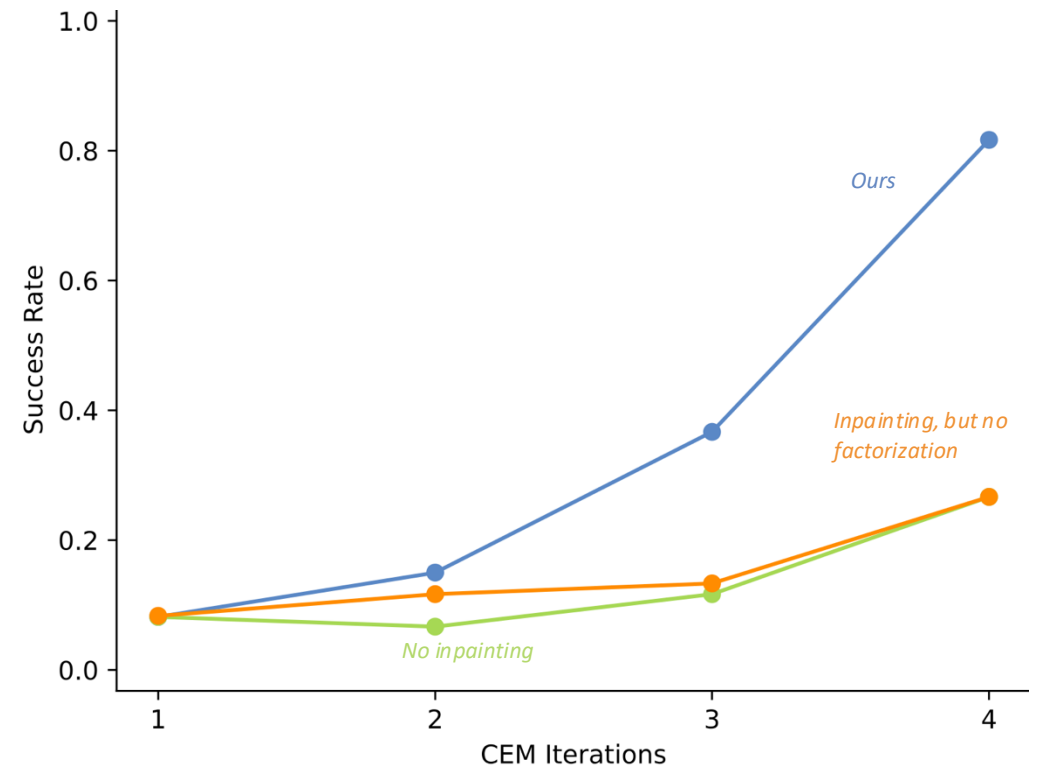
Use it to train robots



Robot Learning through Reward Functions Learned on Video Data



Slide credit: Saurabh Gupta



Affordances from Human Videos as a Versatile Representation for Robotics

Shikhar Bahl^{*1,2}

Russell Mendonca^{*1}

Lili Chen¹

Unnat Jain^{1,2}

Deepak Pathak¹

¹CMU

²Meta AI



Figure 1. We leverage human videos to learn visual affordances that can be deployed on multiple real robot, in the wild, spanning several tasks and learning paradigms. Videos available at <https://vision-robotics-bridge.github.io/>.

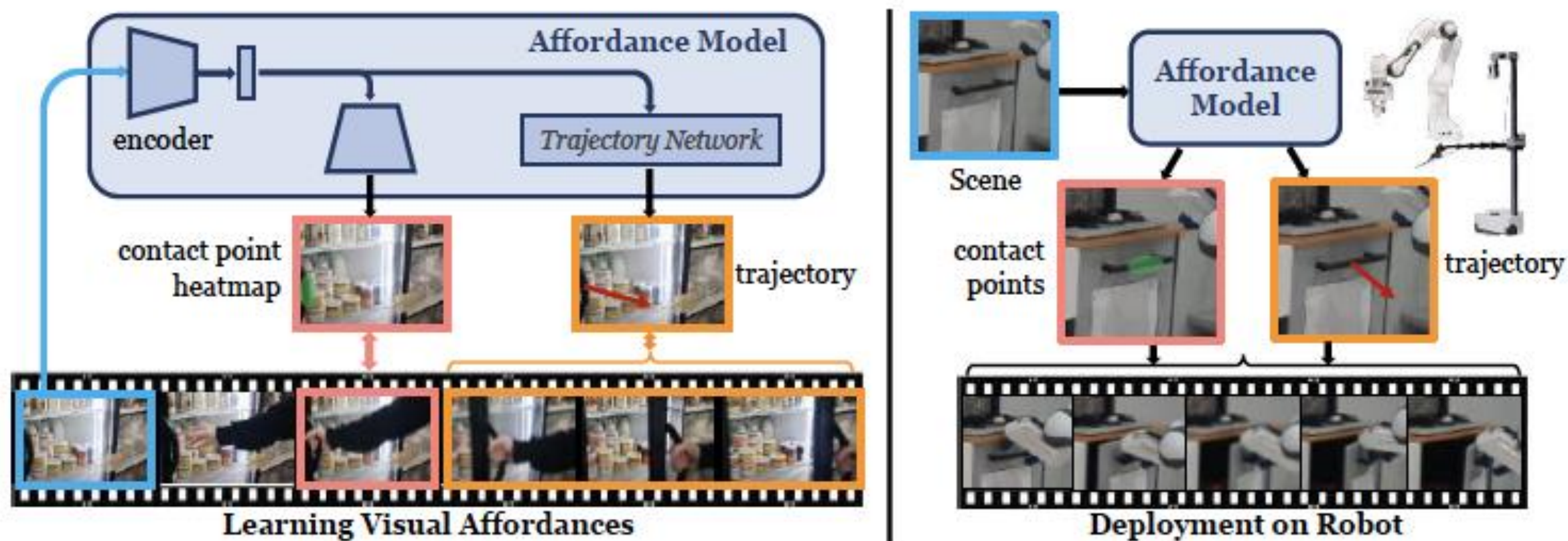


Figure 2. **VRB Overview.** First, we learn an actionable representation of visual affordances from human videos: the model predicts contact points and trajectory waypoints with supervision from future frames. For robot deployment, we query the affordance model and convert its outputs to 3D actions to execute.

<https://vision-robotics-bridge.github.io/>

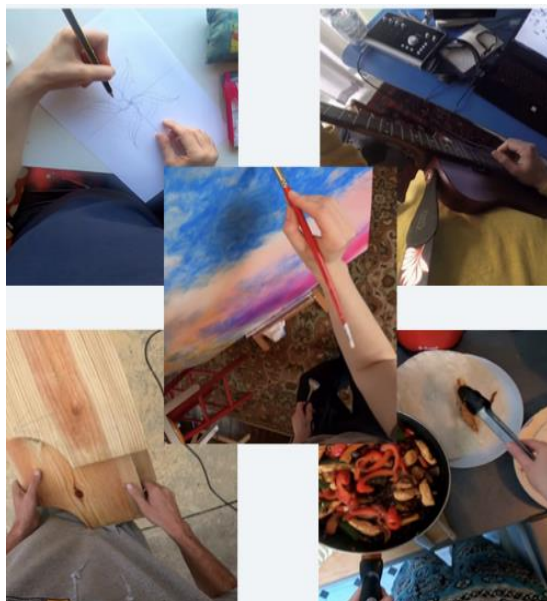
Reconstructing 4D Hand-Object Interactions from Videos

Hand-Object Interaction Pretraining from Videos

Himanshu Gaurav Singh*, Antonio Loquercio*,
Carlo Sferrazza, Jane Wu, Haozhi Qi, Pieter Abbeel, Jitendra Malik

Robot data from videos

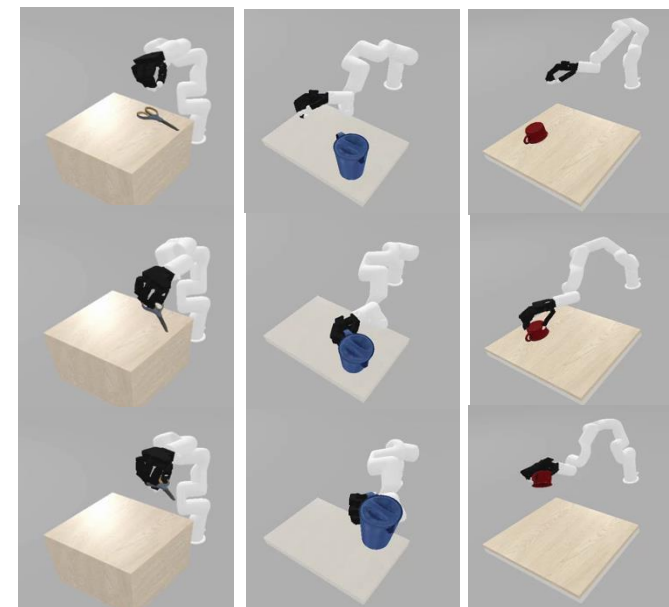
Video Dataset



3-D hand object trajectories

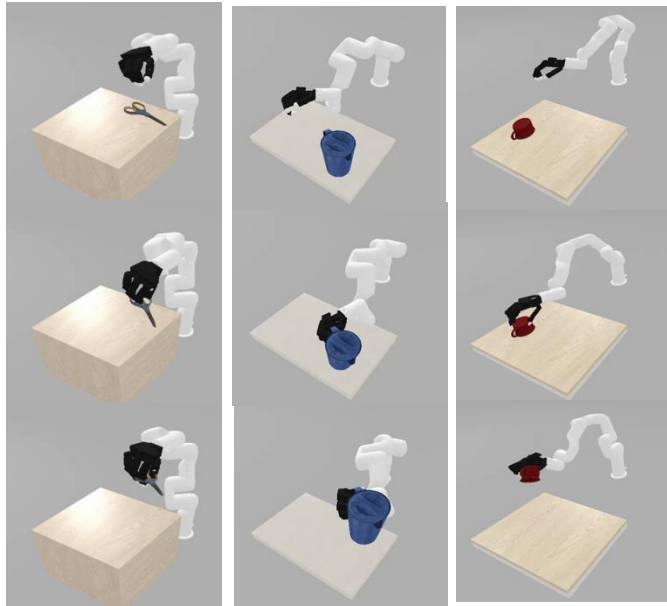


Robot Dataset

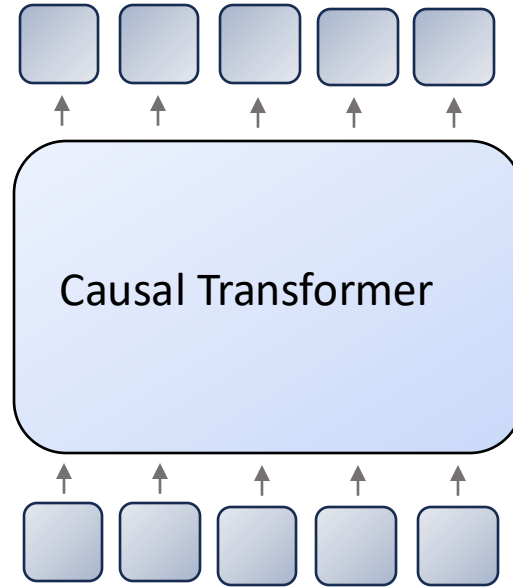


Learning robot policies

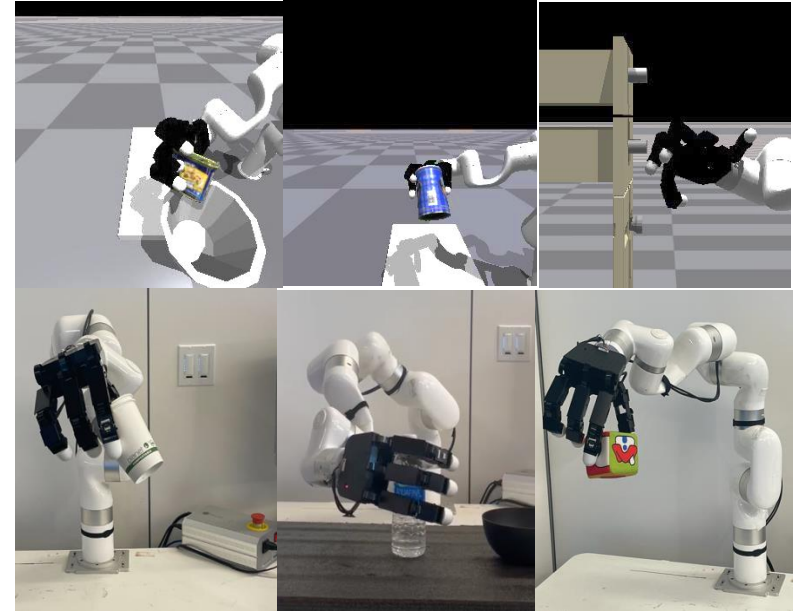
Robot Dataset



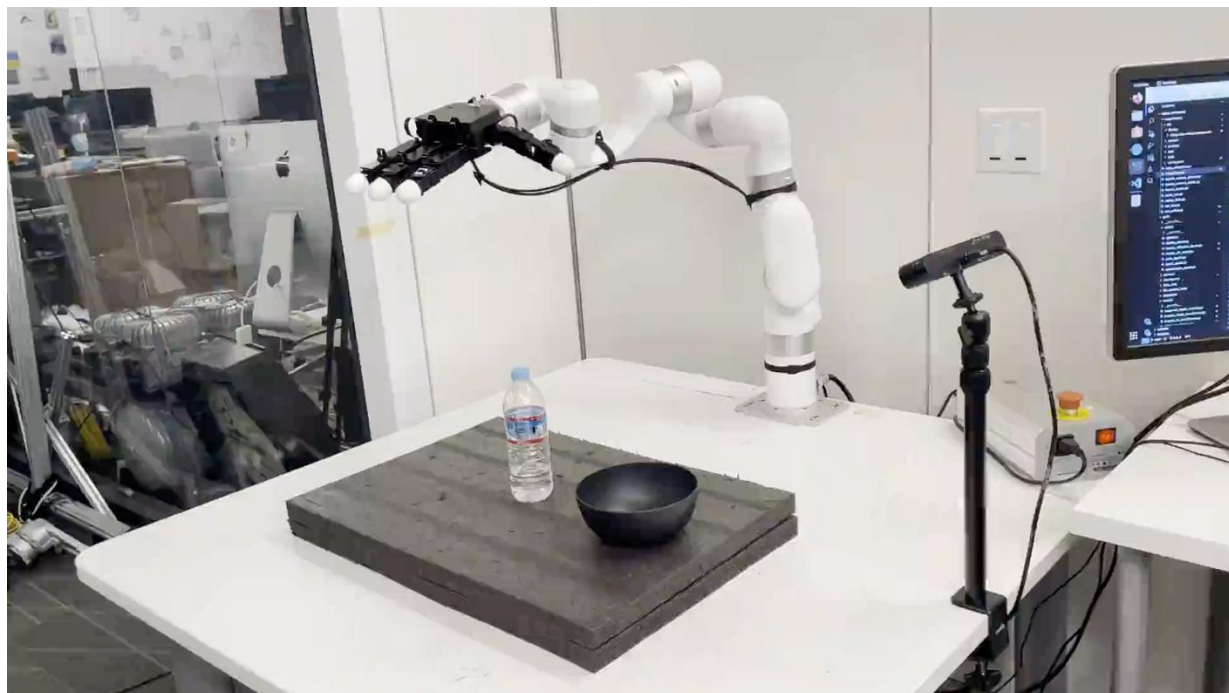
General Manipulation
Prior



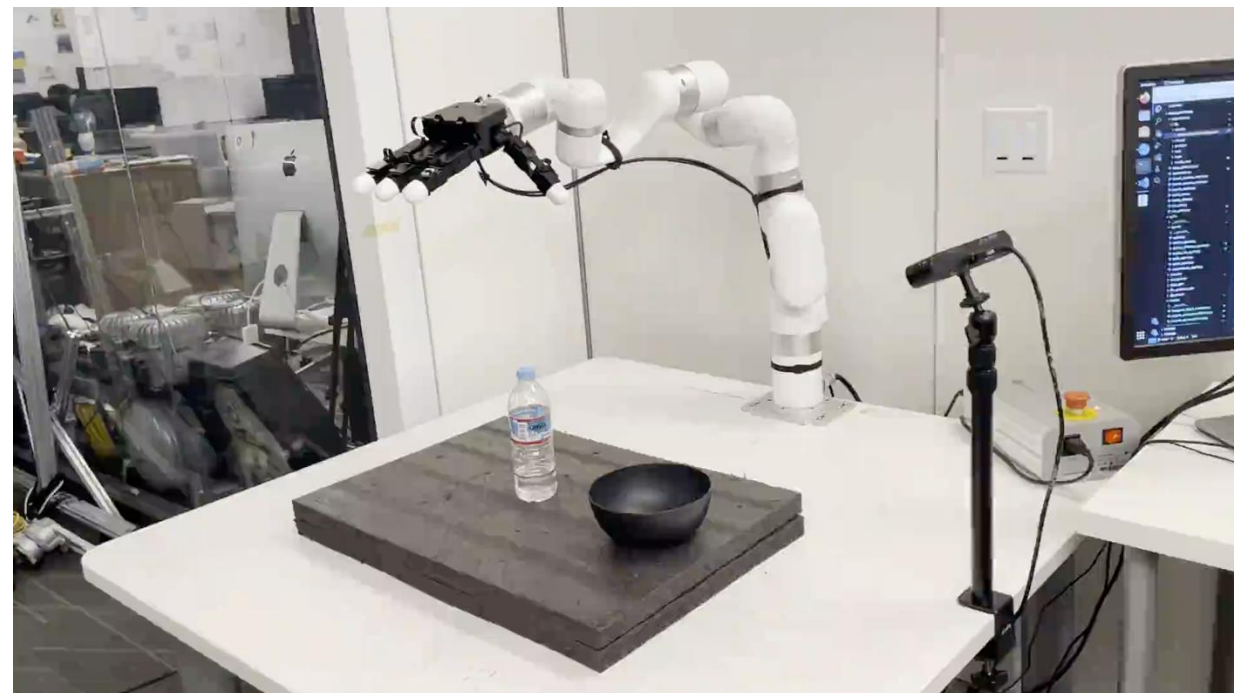
Downstream adaptation



Grasp and Pour



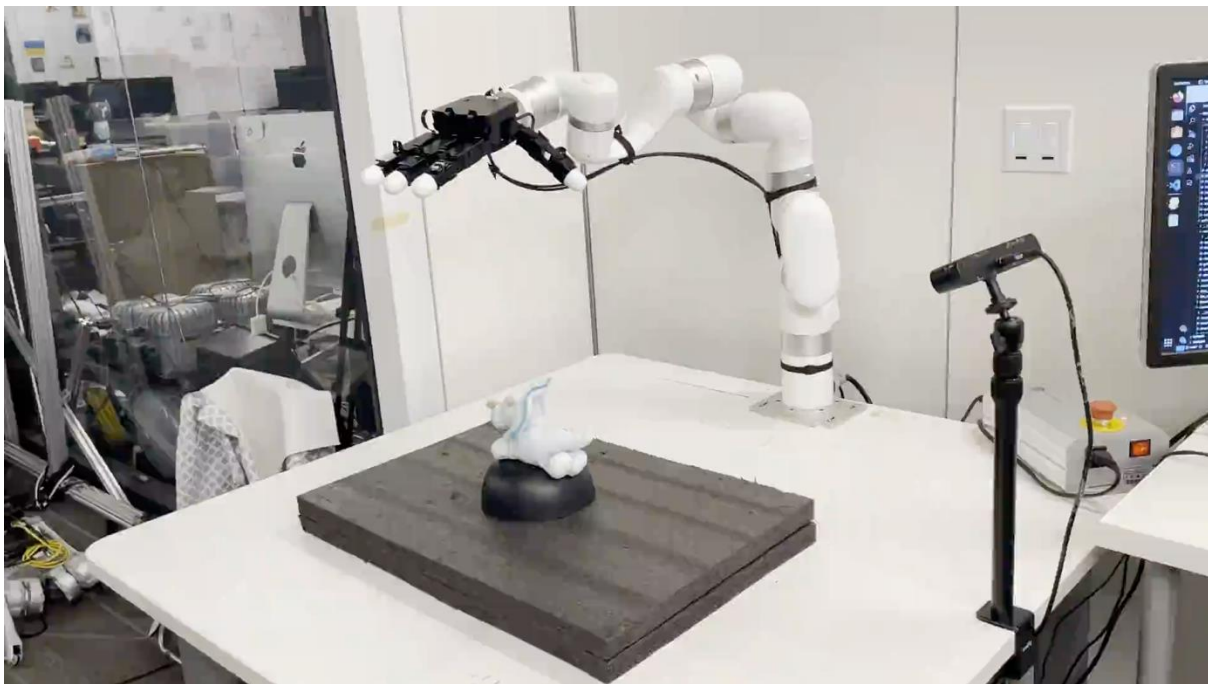
Ours



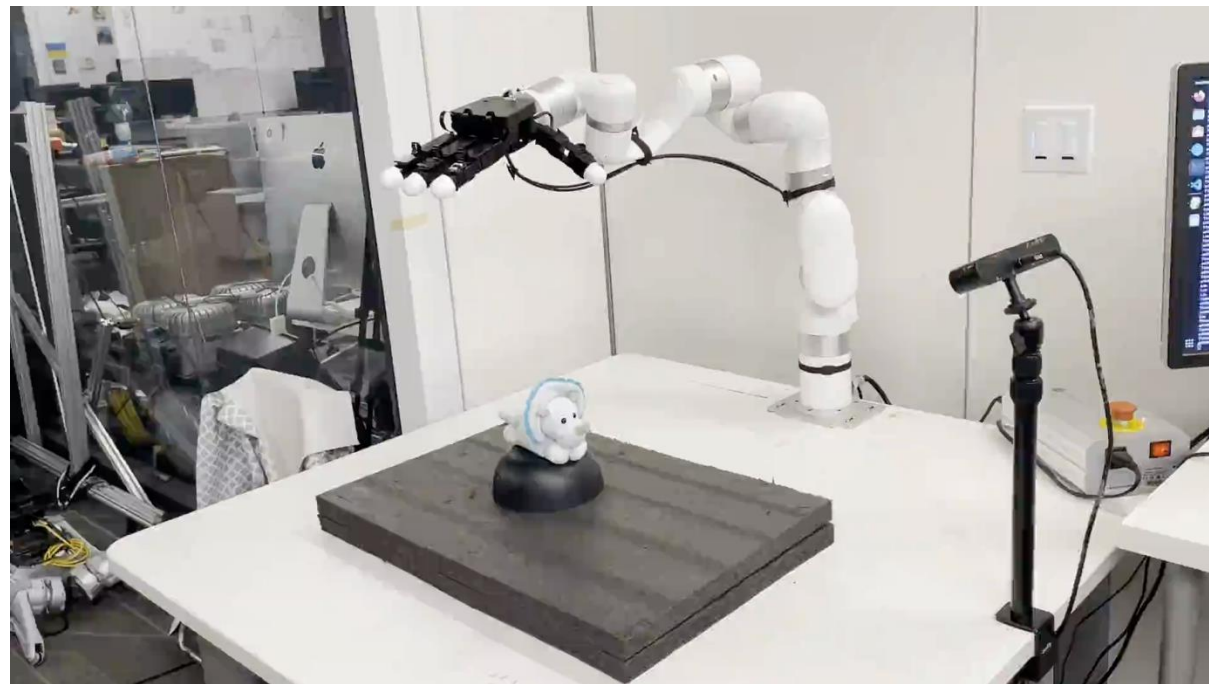
Diffusion Policy
(shaky trajectory)

After downstream finetuning with < 15 demonstrations

Grasp and Lift (toy)



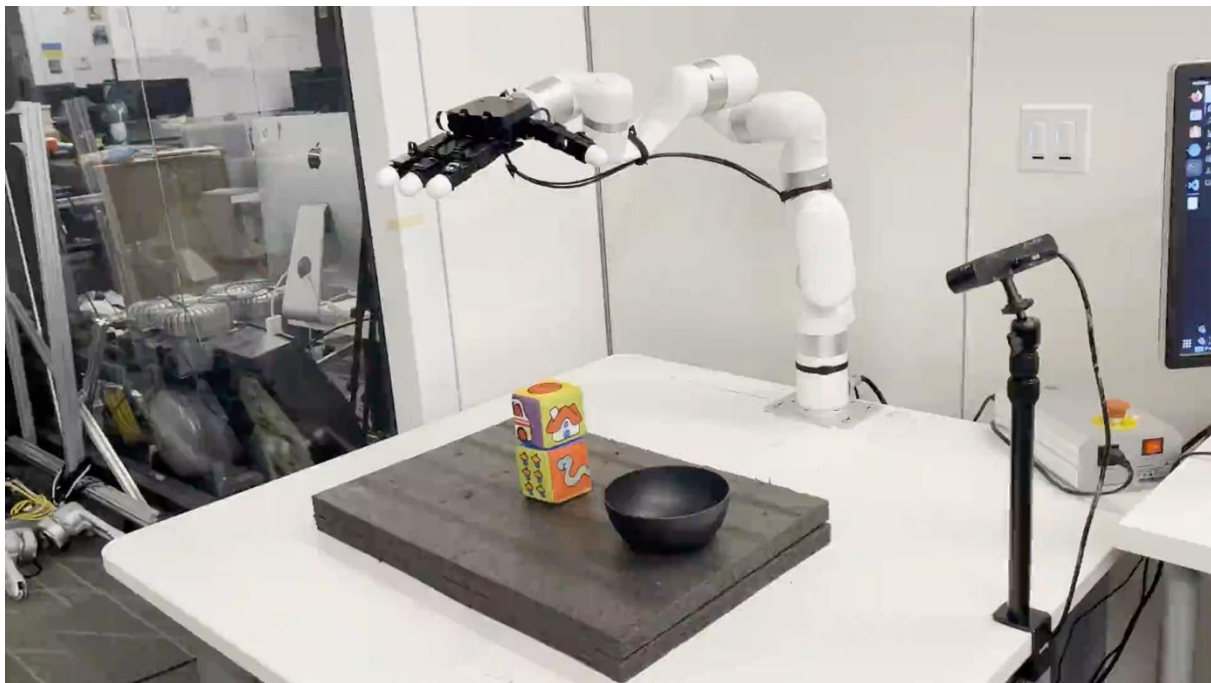
Ours



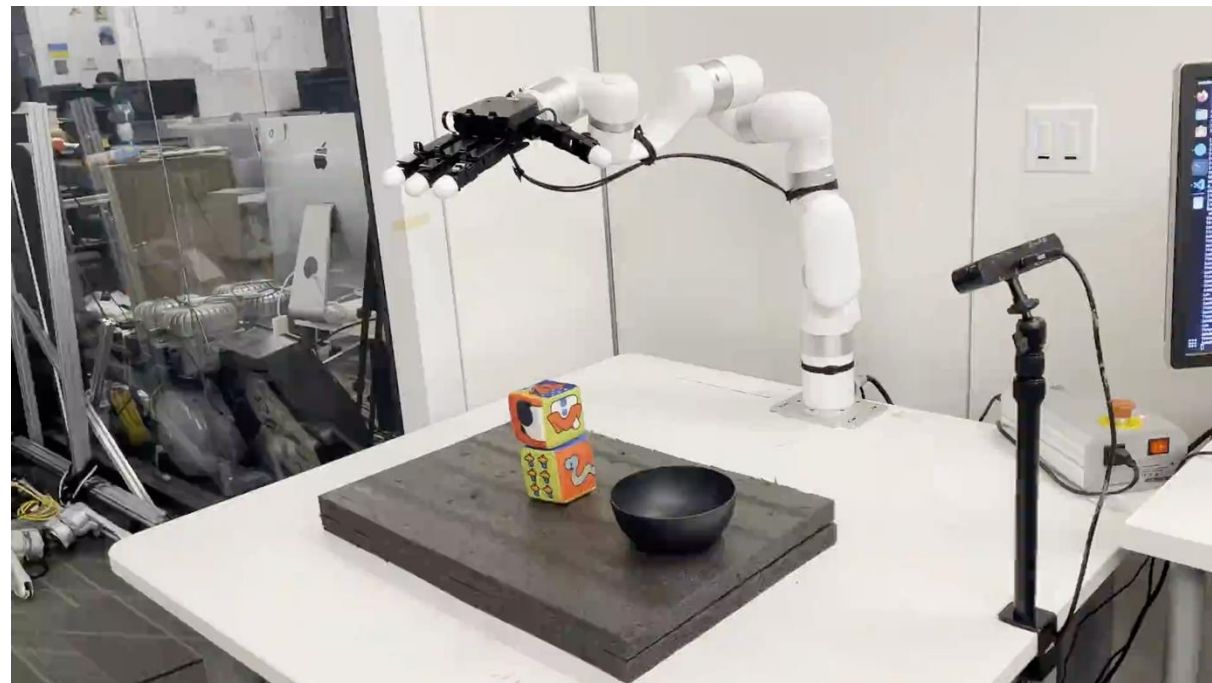
Diffusion Policy
(shaky trajectory)

After downstream finetuning

Grasp and Throw (cube)



Ours



Diffusion Policy
(shaky behavior)

After downstream adaptation with < 15 demonstrations

Thank you!